

Sample analysis

April 13, 2025

```
[8]: %reload_ext autoreload
      %autoreload 2
      from experiment import *

      params = (968872.5091425295, 652219.0269692114, 1.81351143628122e-05, 1.
        ↪ 722294477970062e-05, -6.651860692596738e-08, -5.693150723984048e-07)
      data = Active_Analysis('250226/bead3')
```

Calibration parameters:

Sx: 820074.2474561329

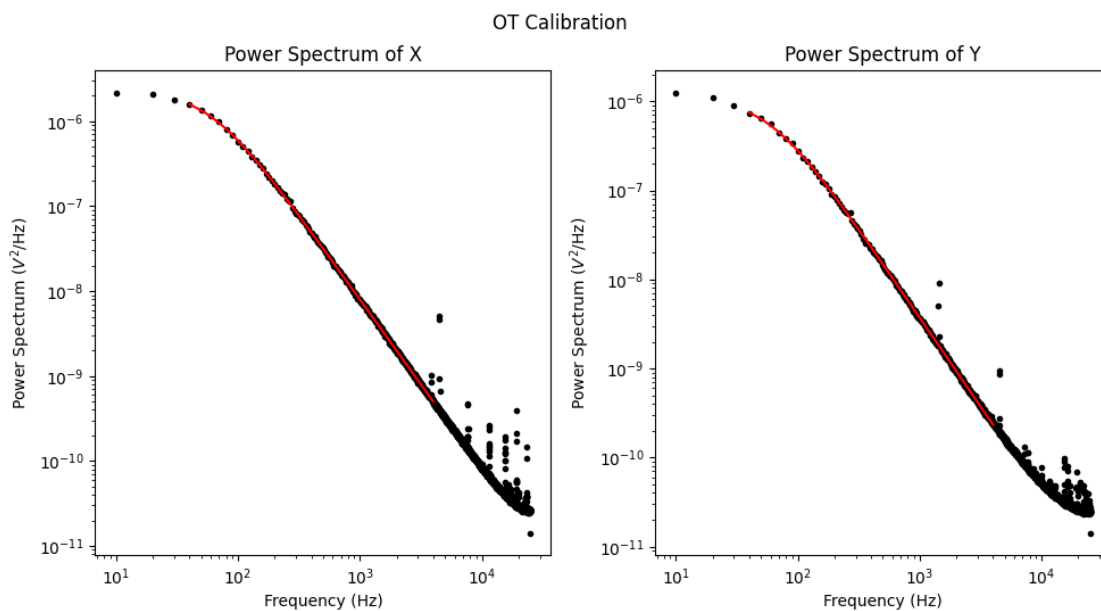
Sy: 560741.2730406355

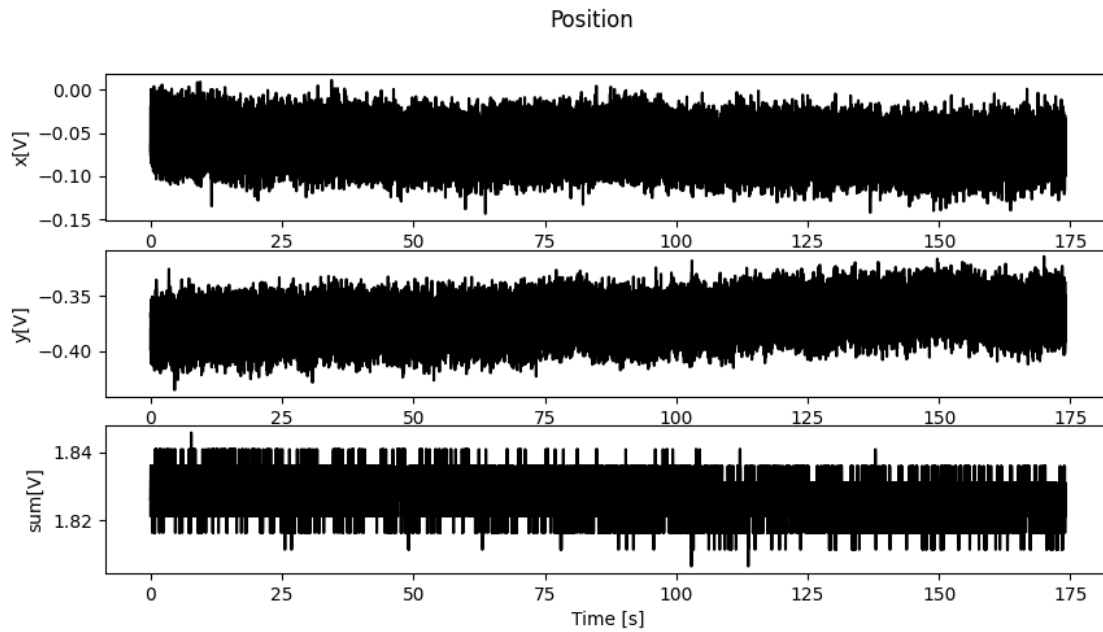
kx: 1.2912493813158651e-05

ky: 1.2801834934988684e-05

trapx: -7.858806662560769e-08

trapy: -6.621915318362614e-07





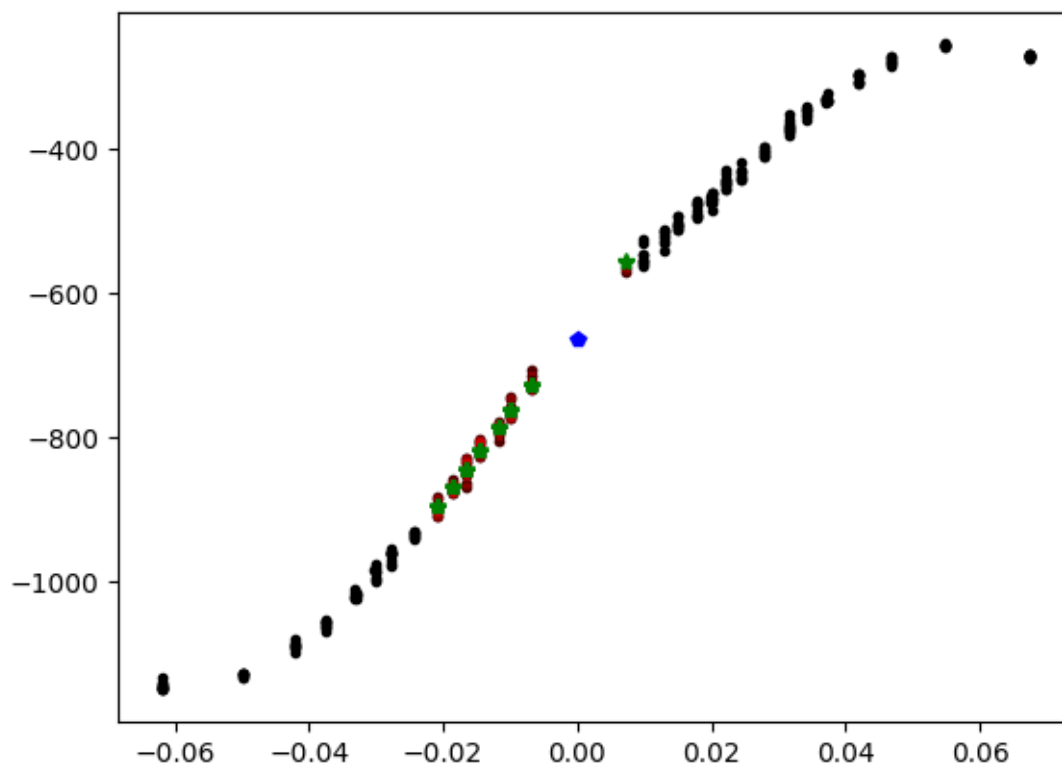
```
[6]: 820074.2474561329/1e6
```

```
[6]: 0.8200742474561329
```

```
[40]: -5.693150723984048e-07/-4.325915129161261e-08
```

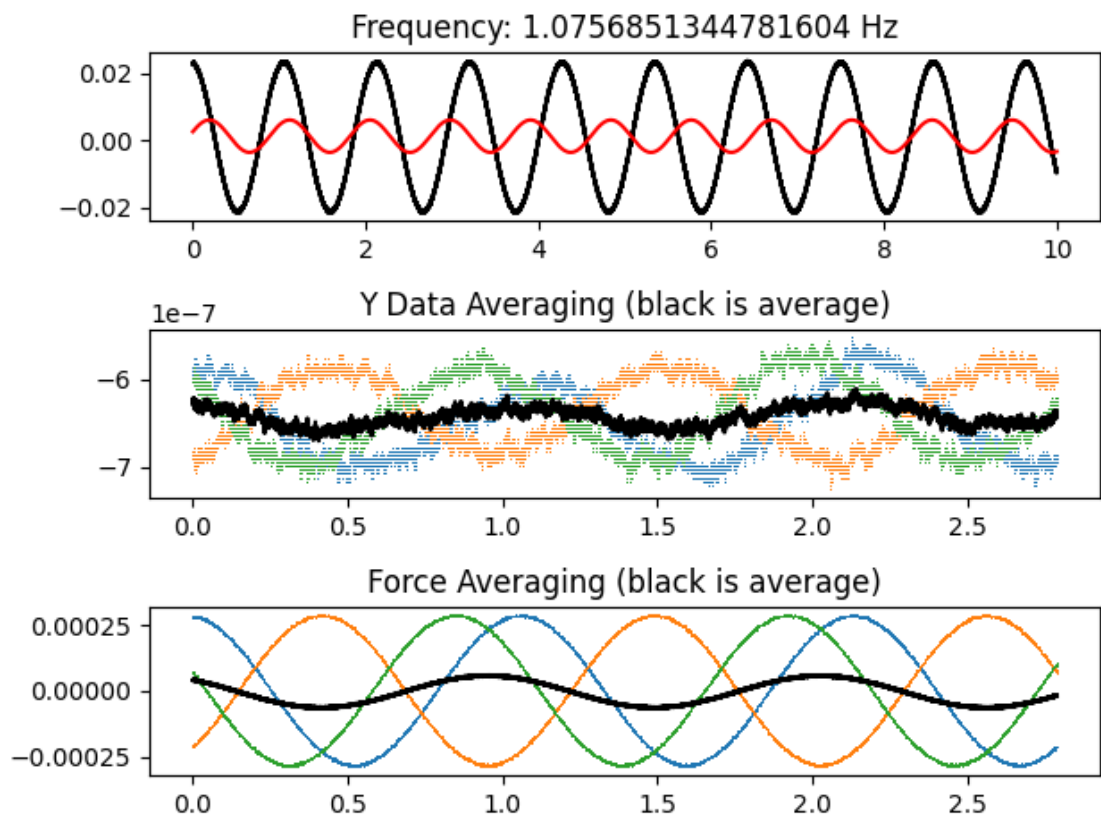
```
[40]: 13.160569622843889
```

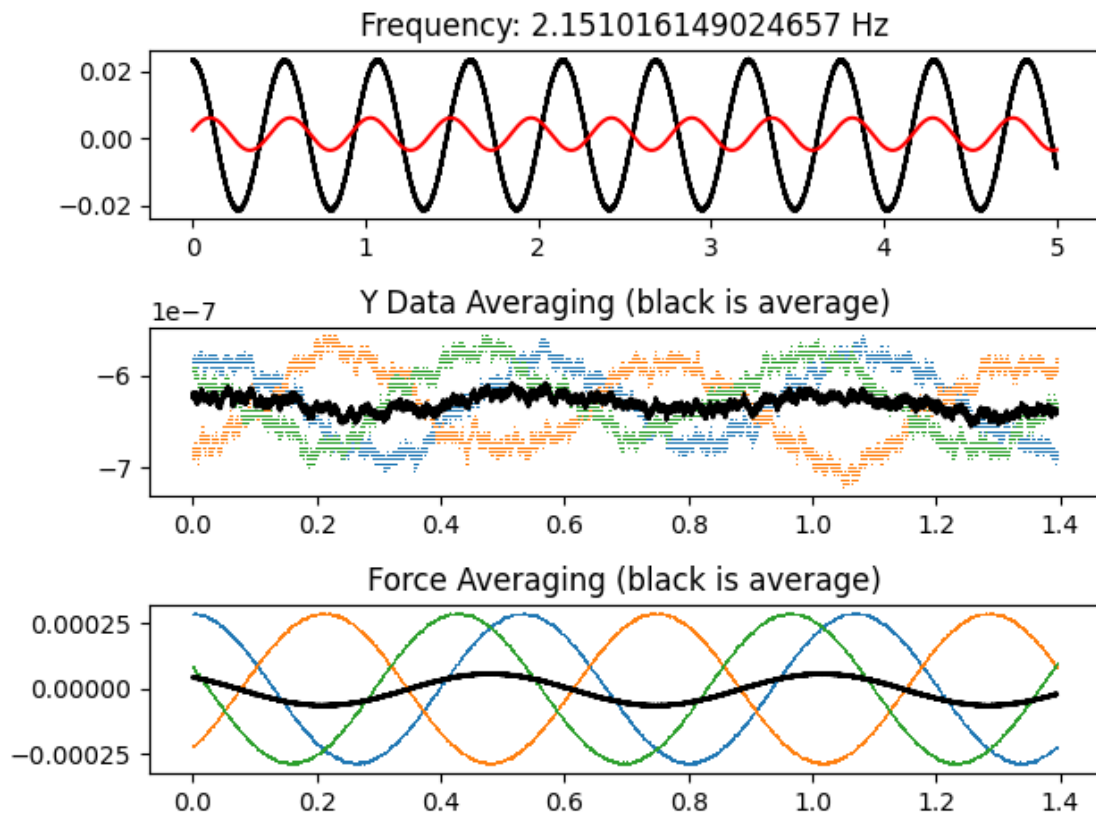
```
[9]: data.calibrate_function_gen([-250e-9, 100e-9])
```

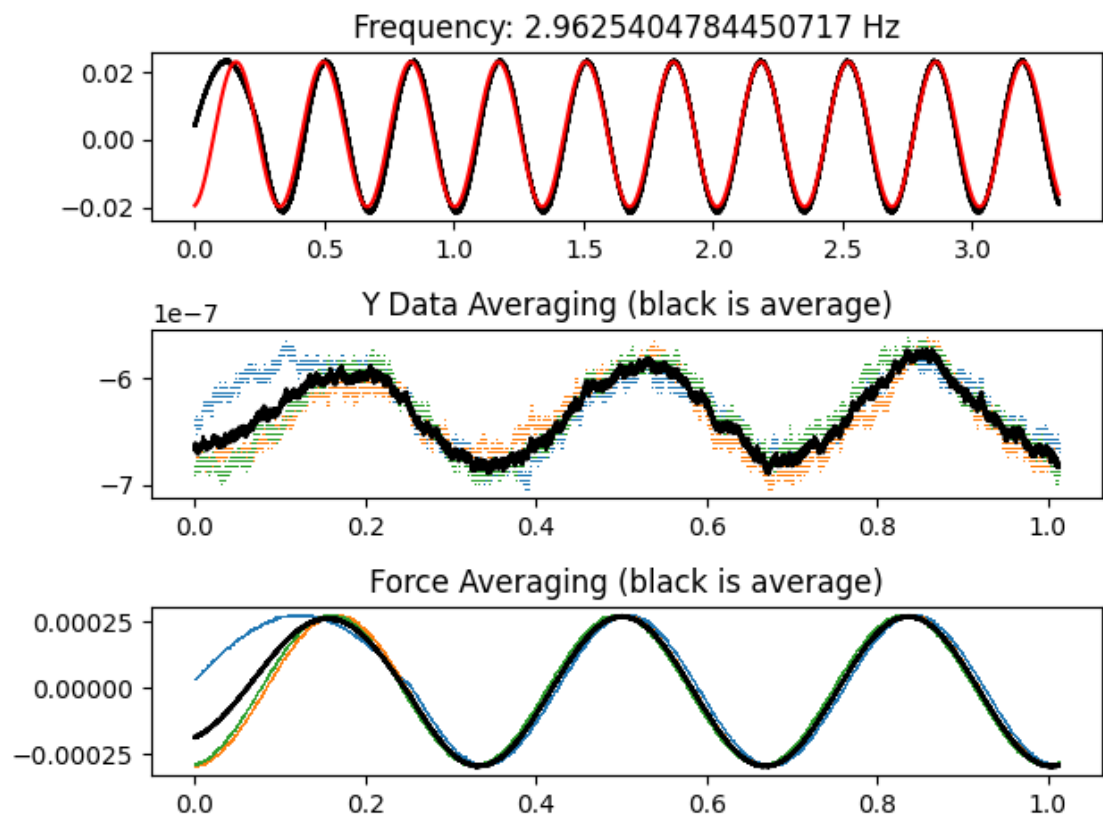


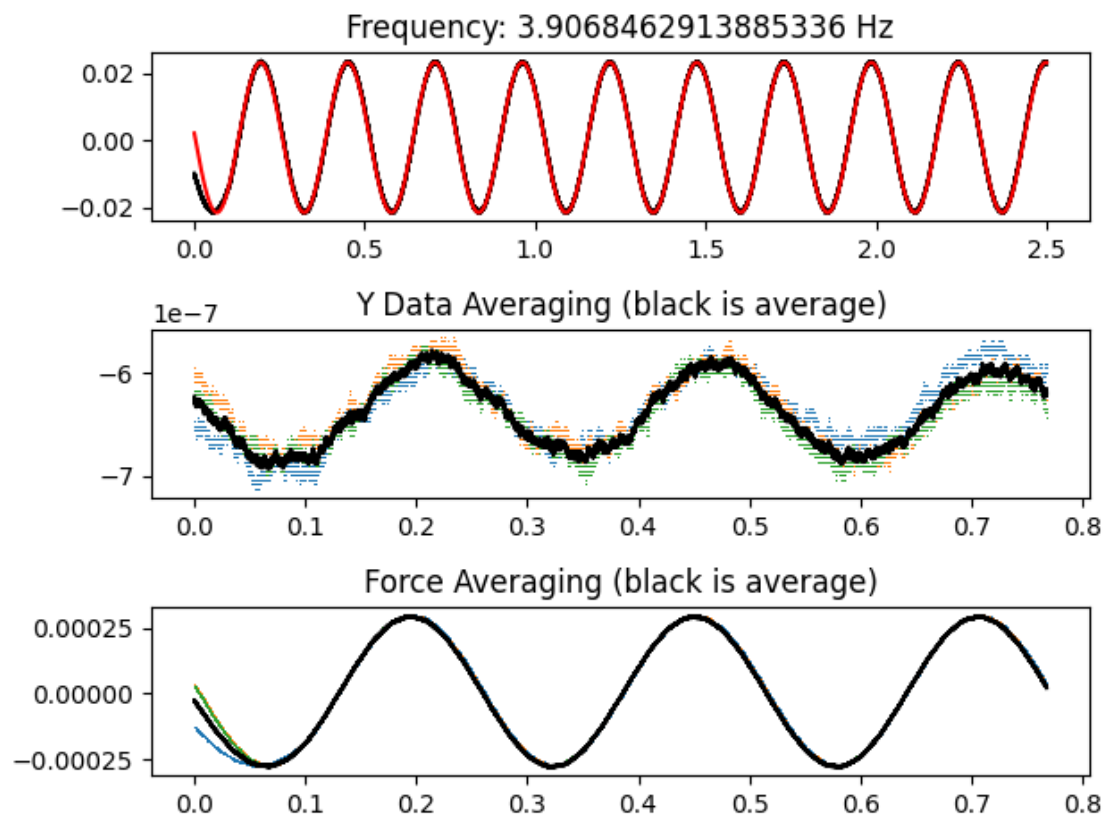
```
[19]: data.final_analysis()
```

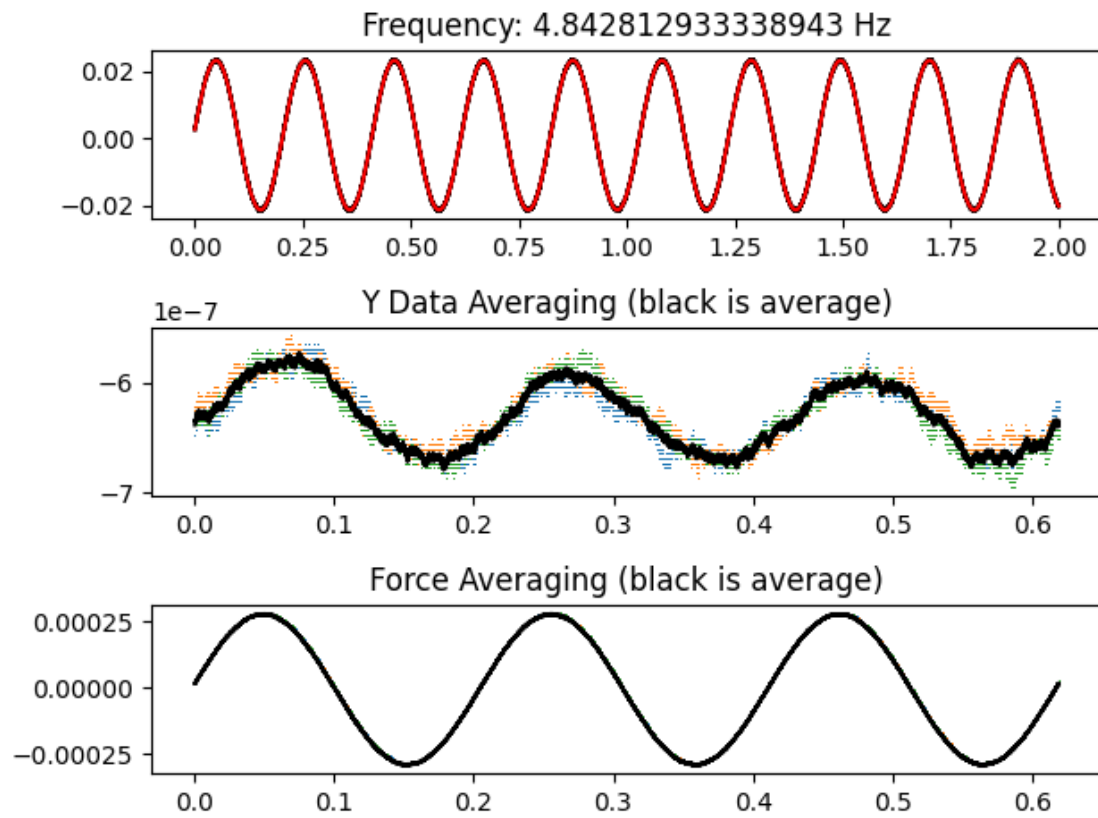
/Users/colededebois/Altman lab/Altmanlab/experiment.py:554: RuntimeWarning: More than 20 figures have been opened. Figures created through the pyplot interface (`matplotlib.pyplot.figure`) are retained until explicitly closed and may consume too much memory. (To control this warning, see the rcParam `figure.max_open_warning`). Consider using `matplotlib.pyplot.close()`.
 plt.figure(tight_layout=True)

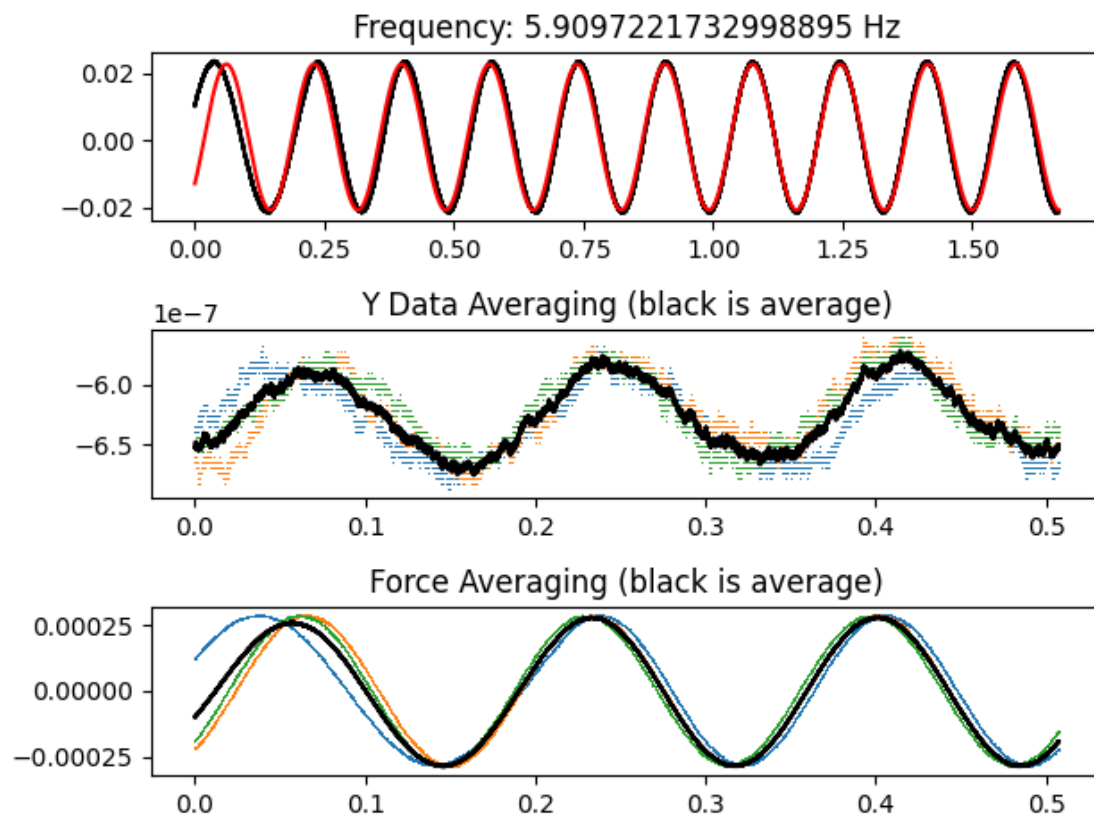


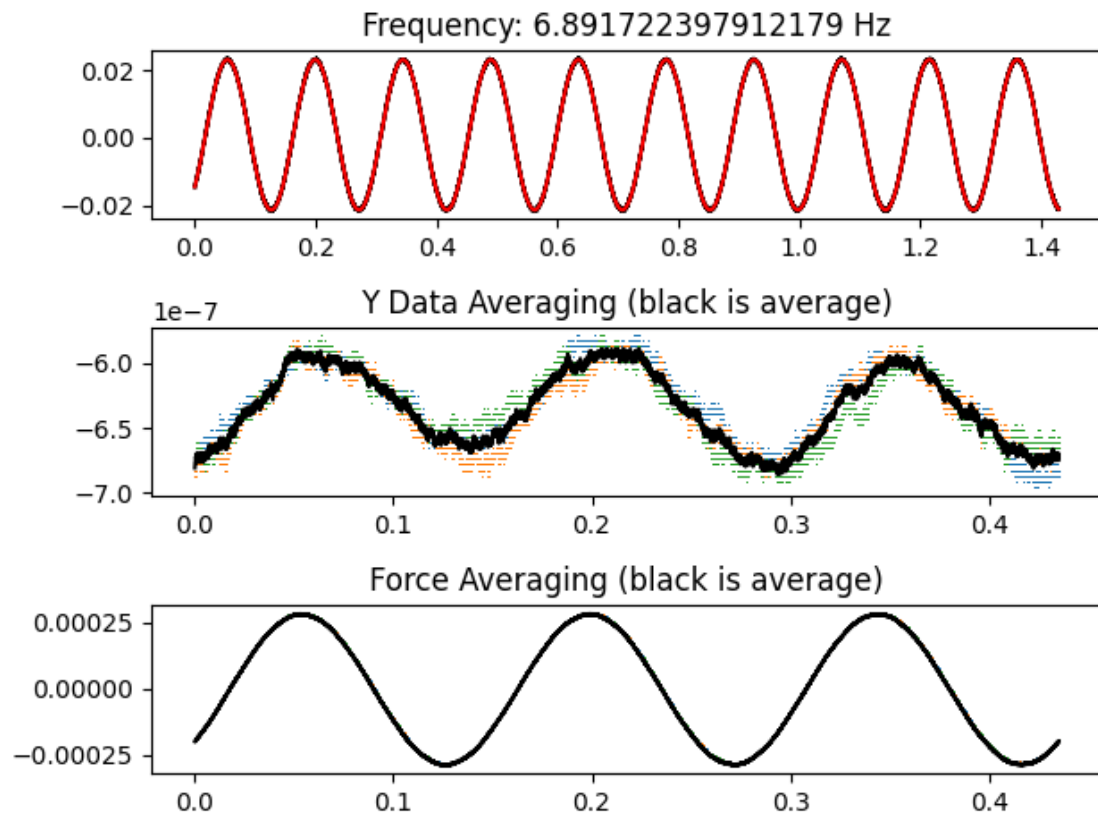


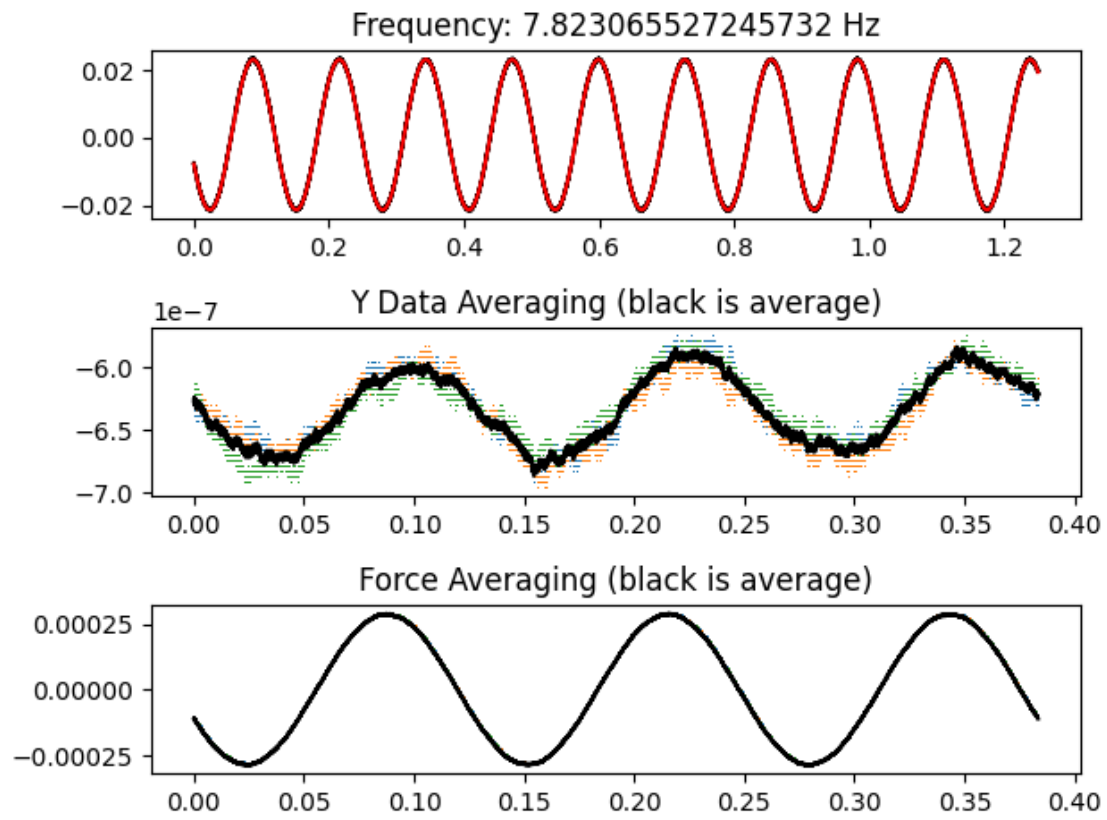


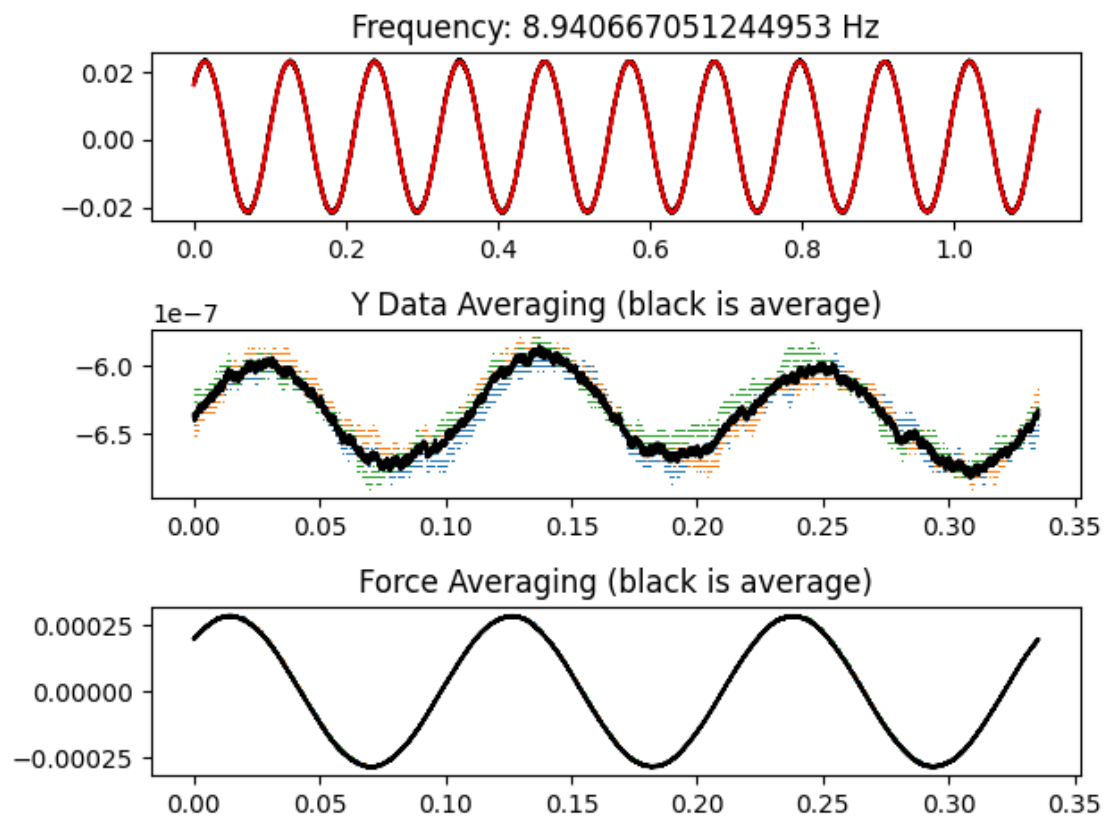


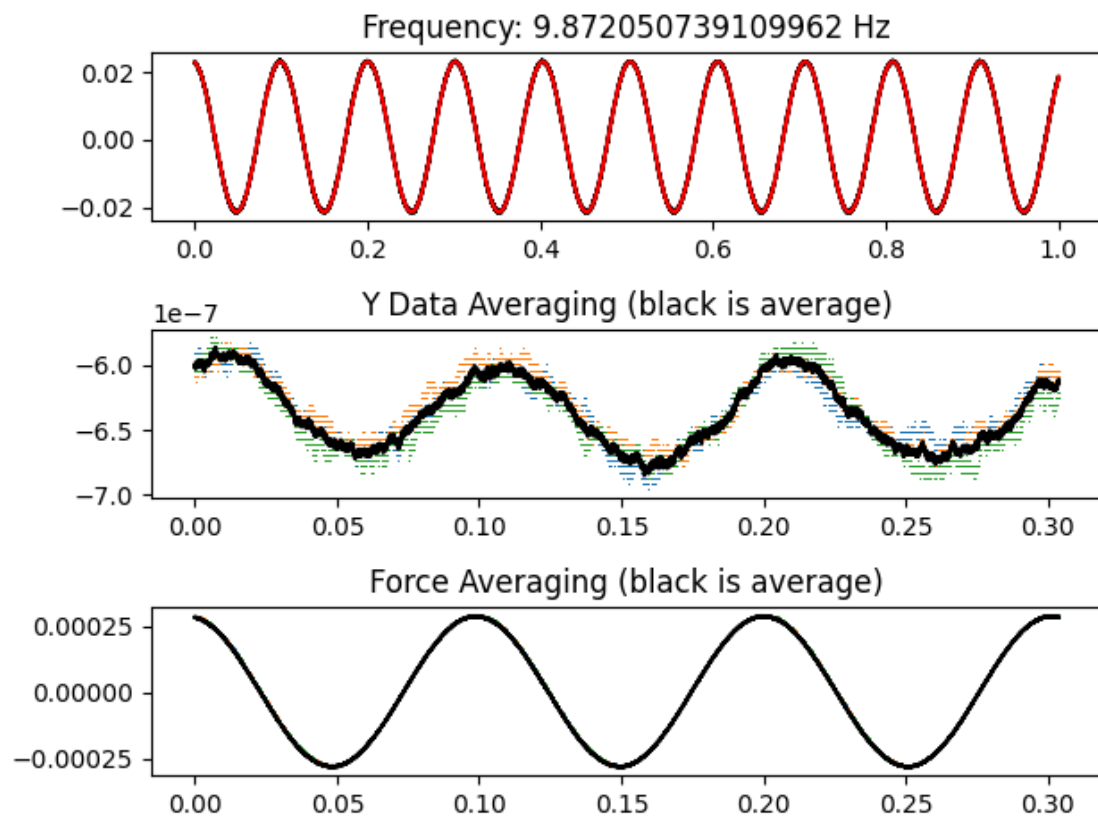


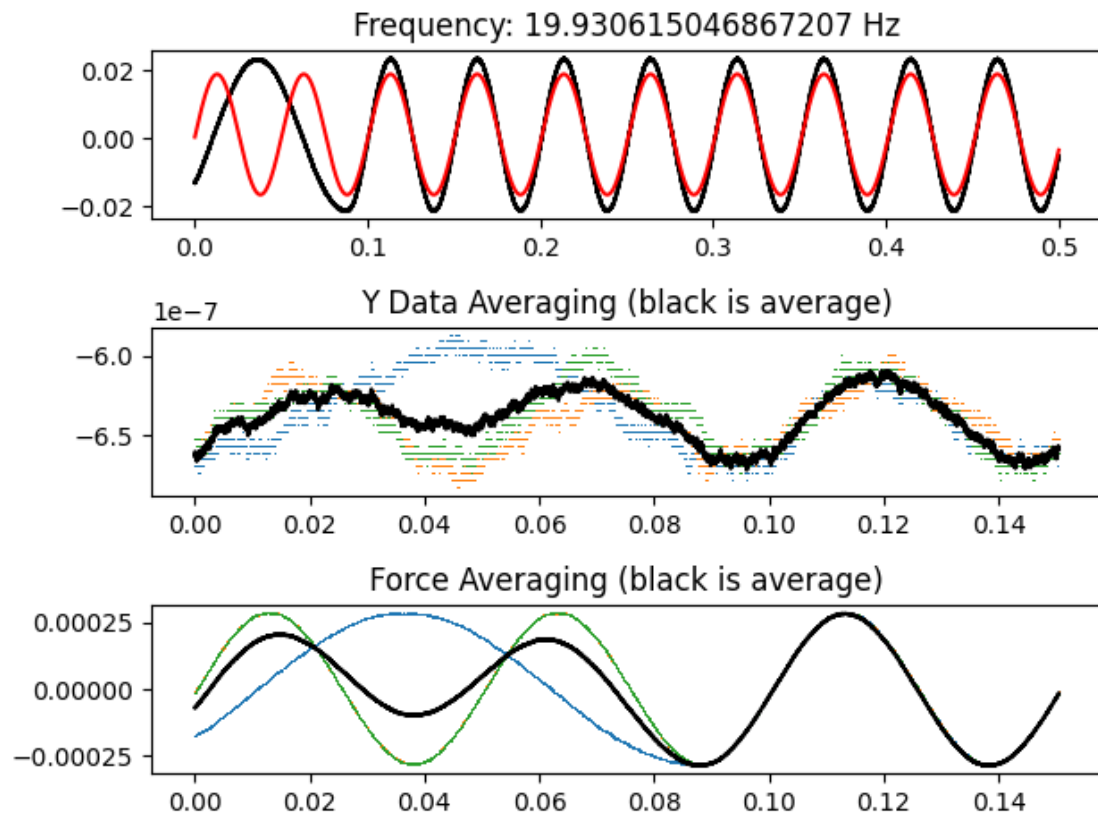


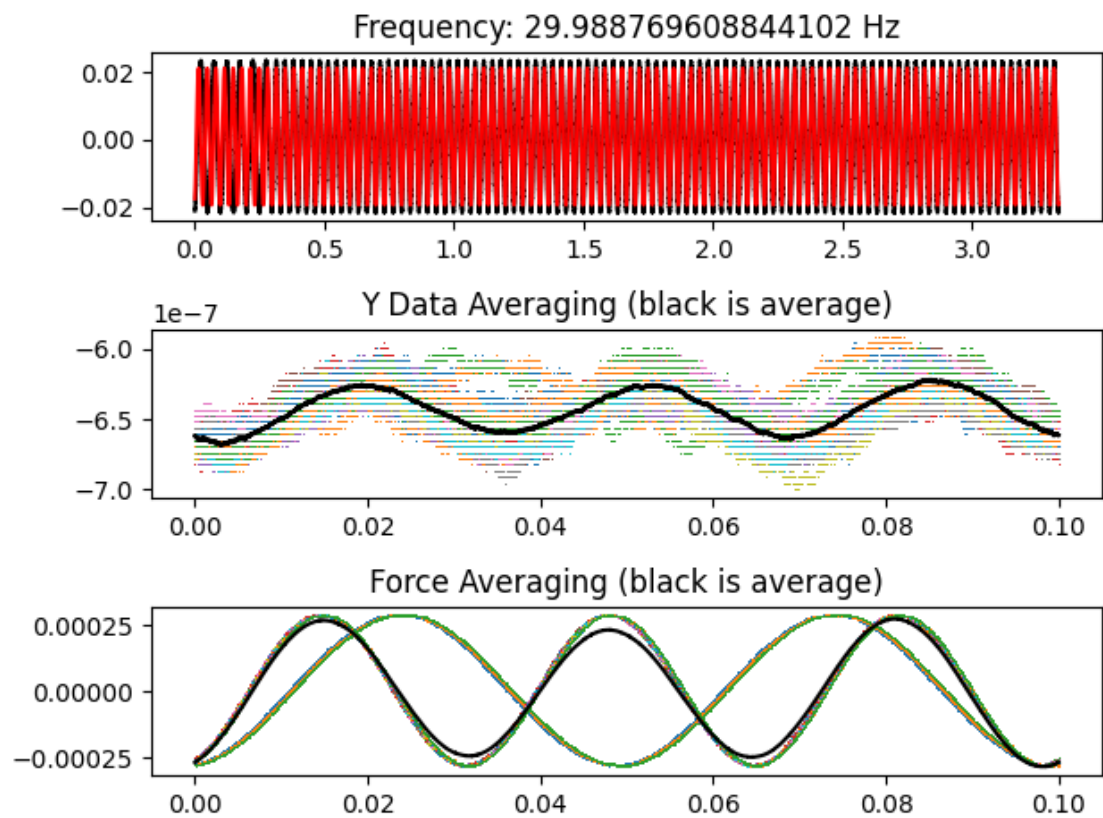


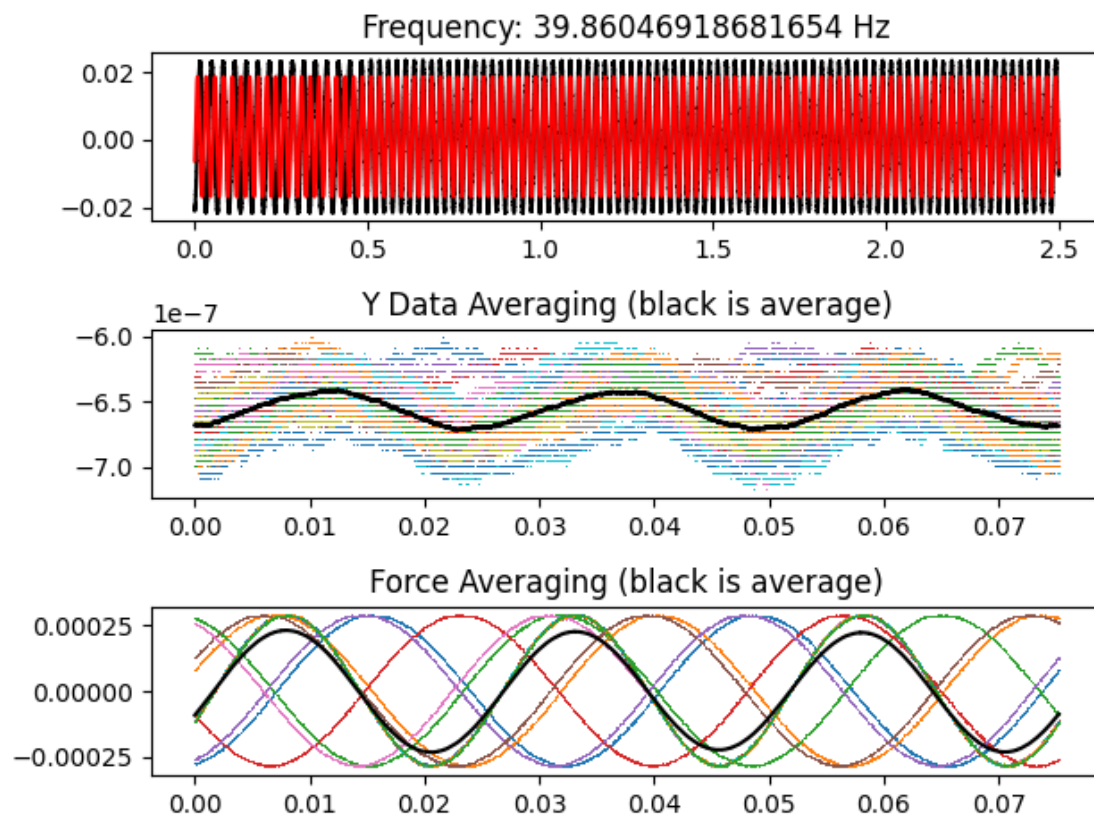


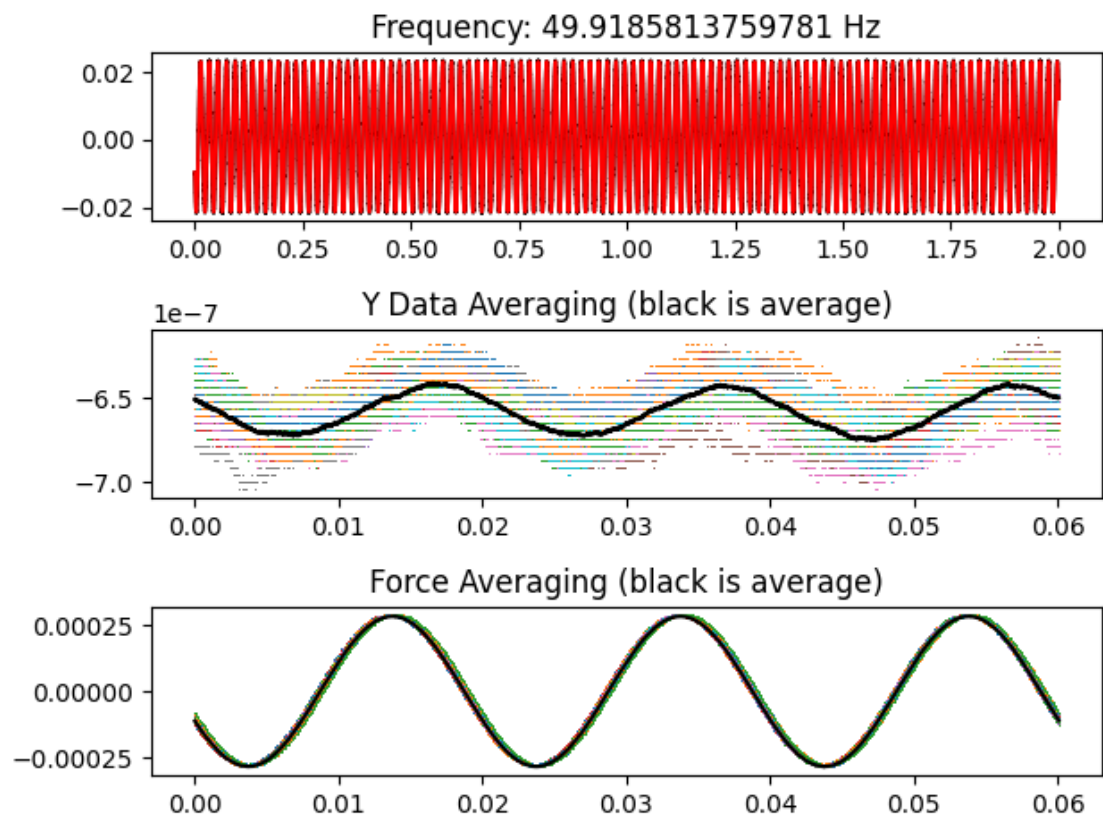


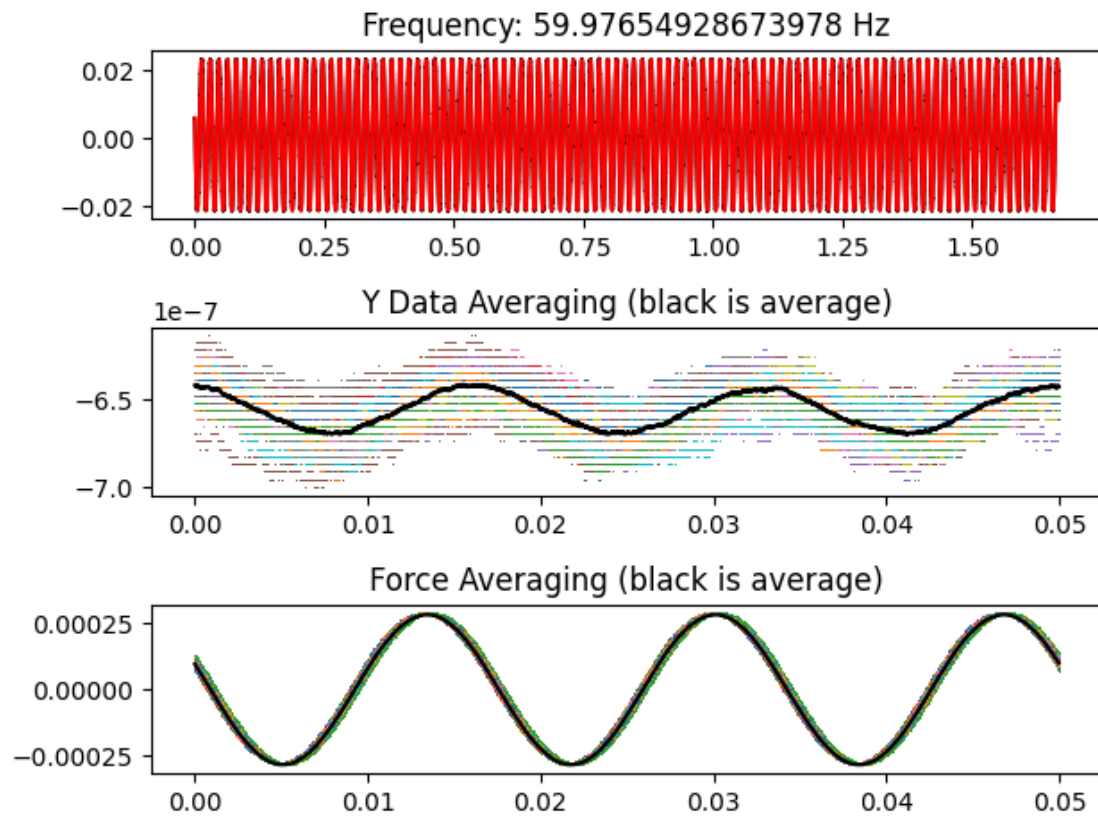


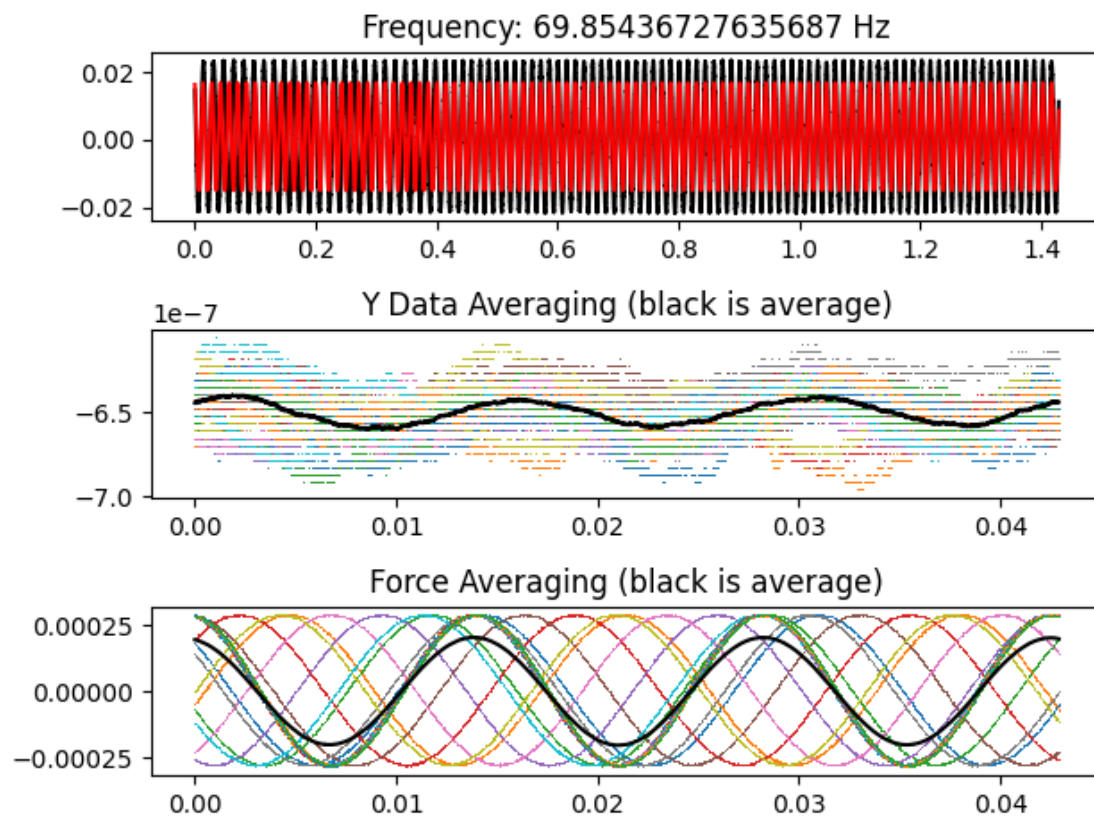


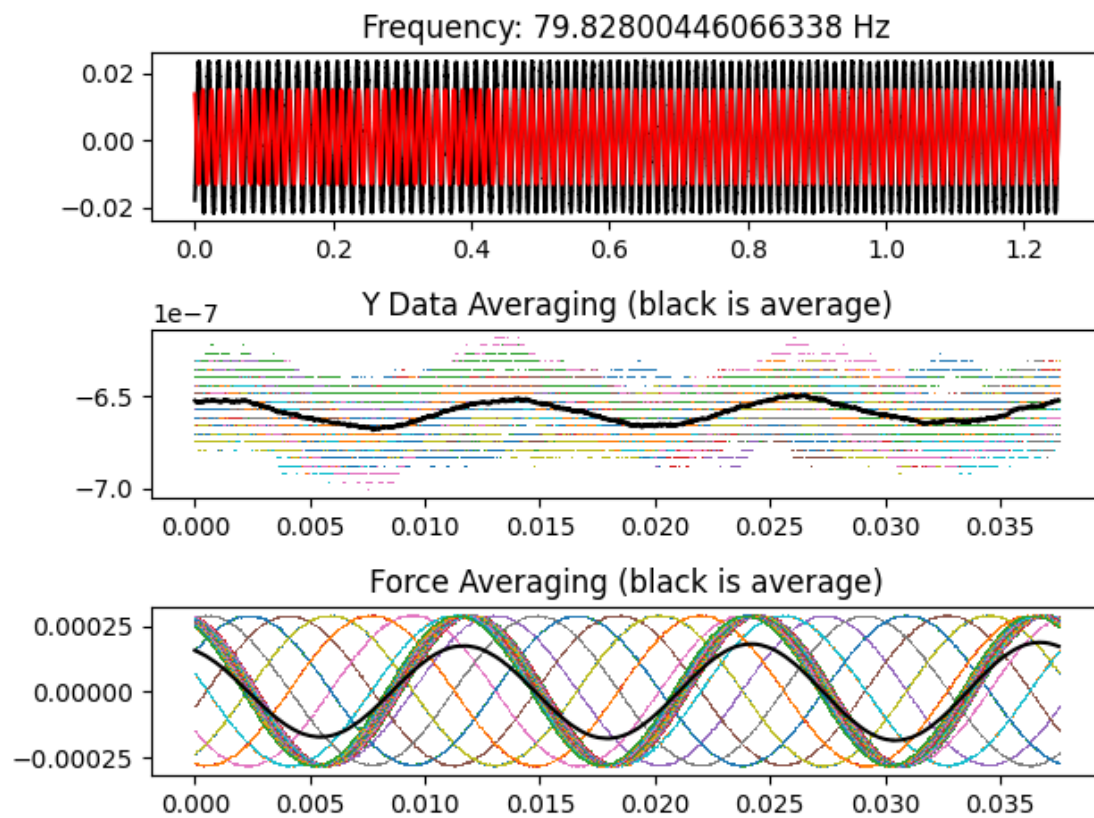


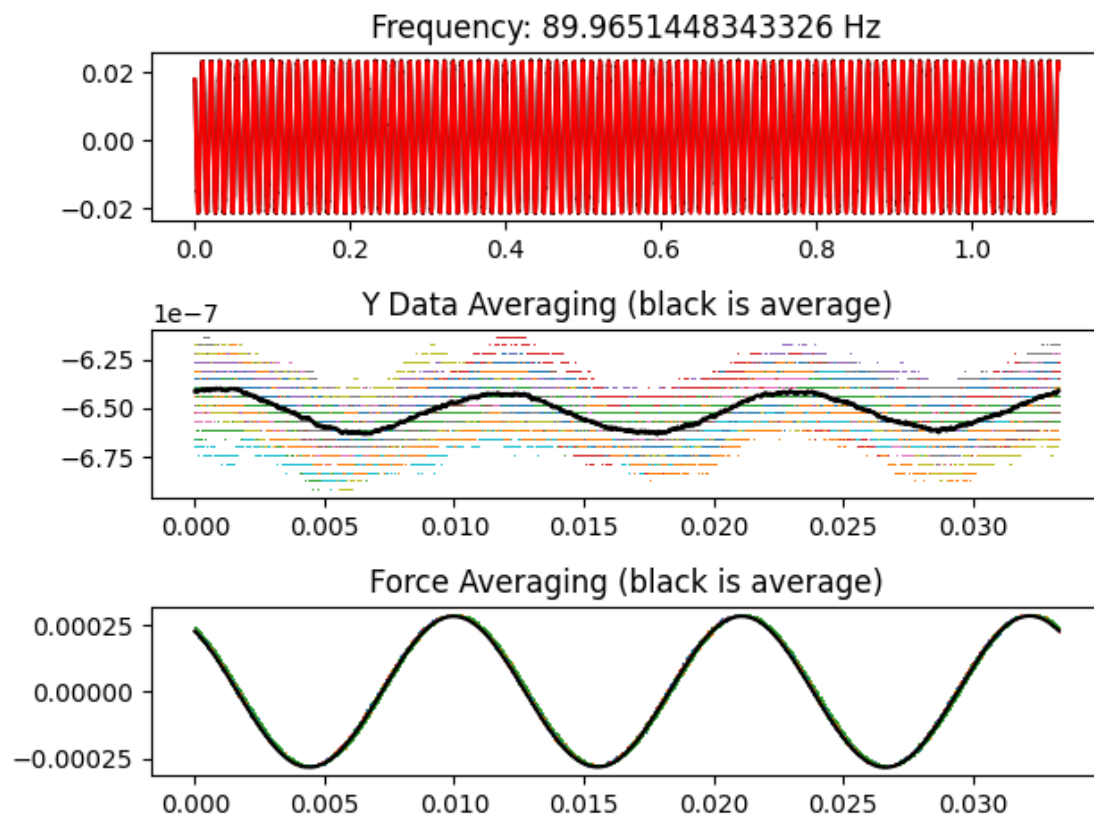


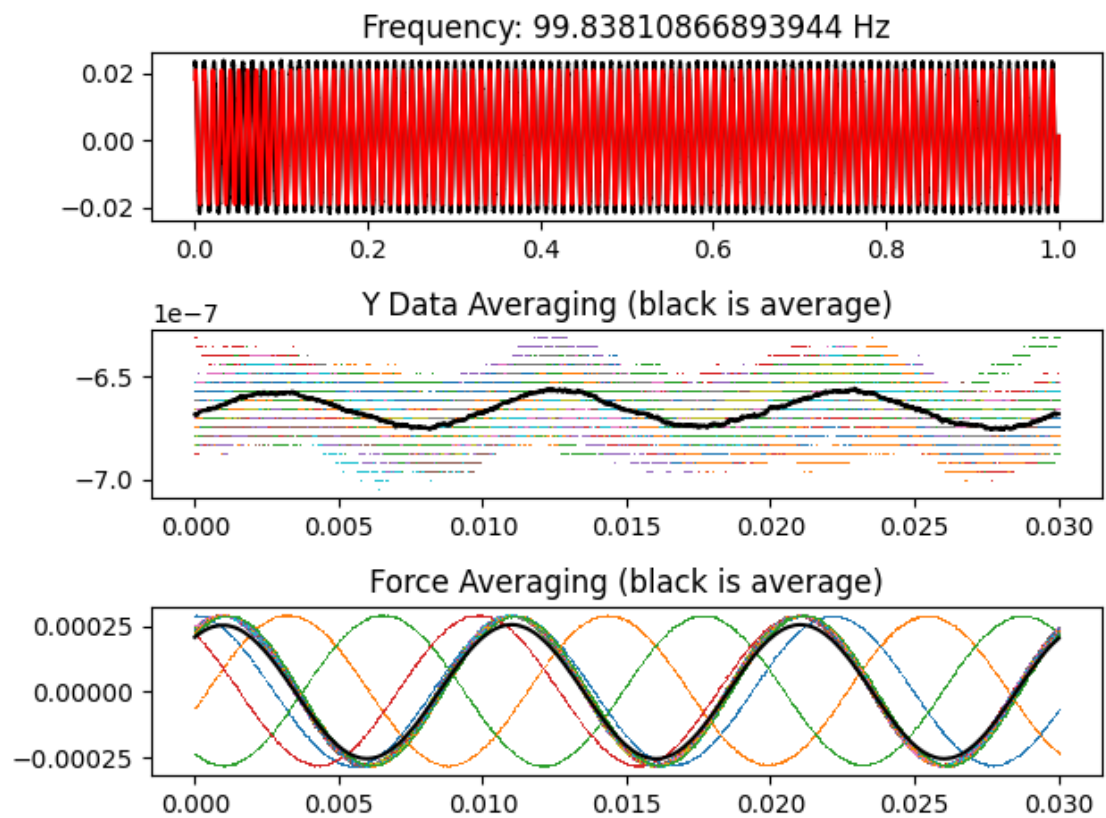


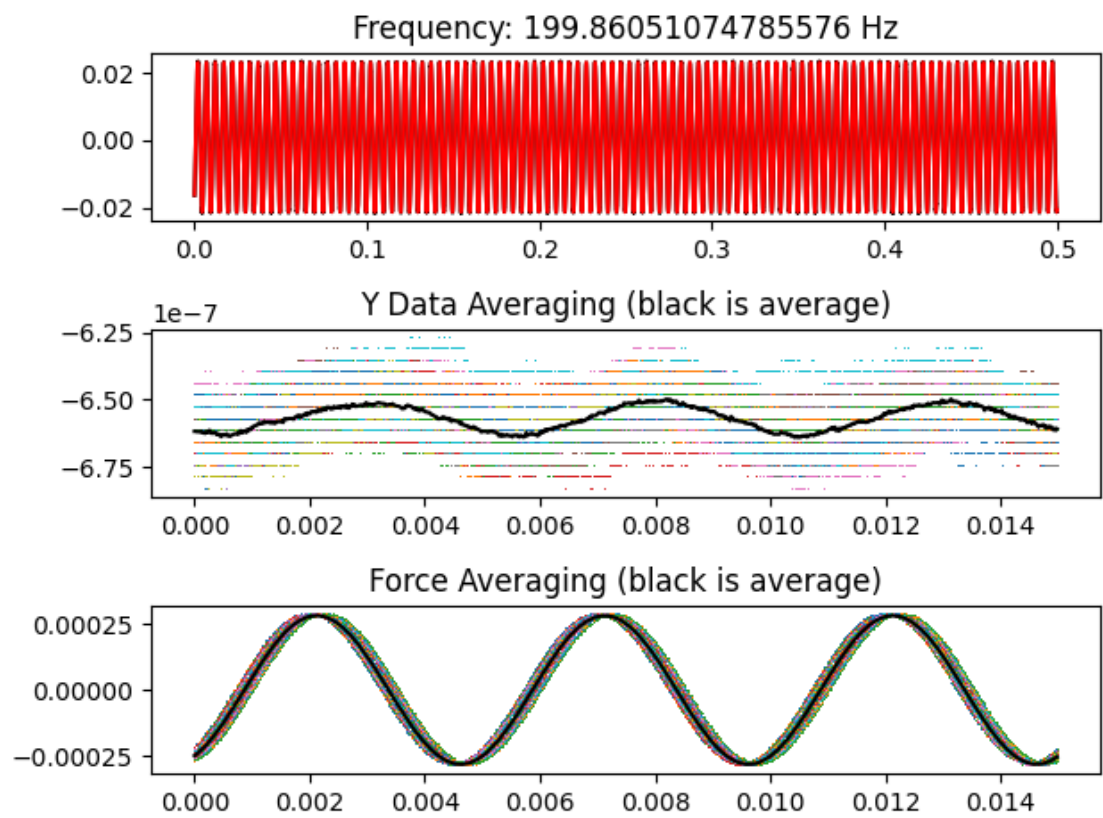


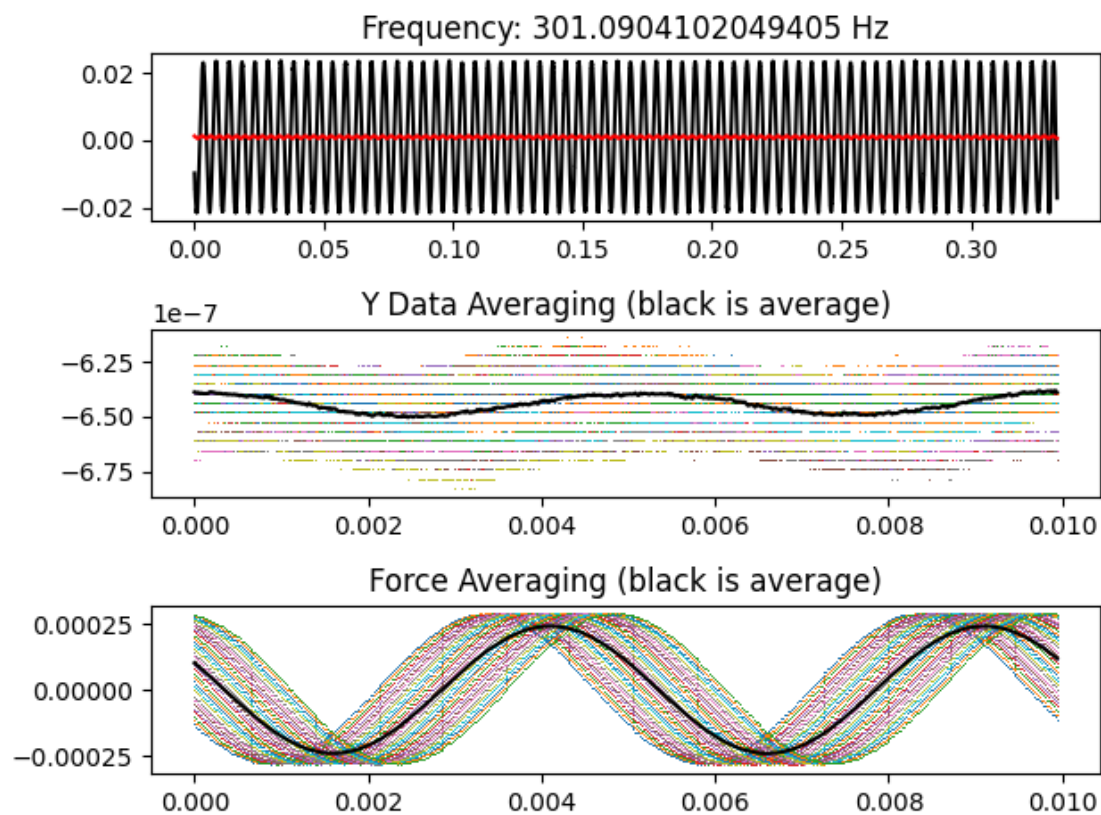


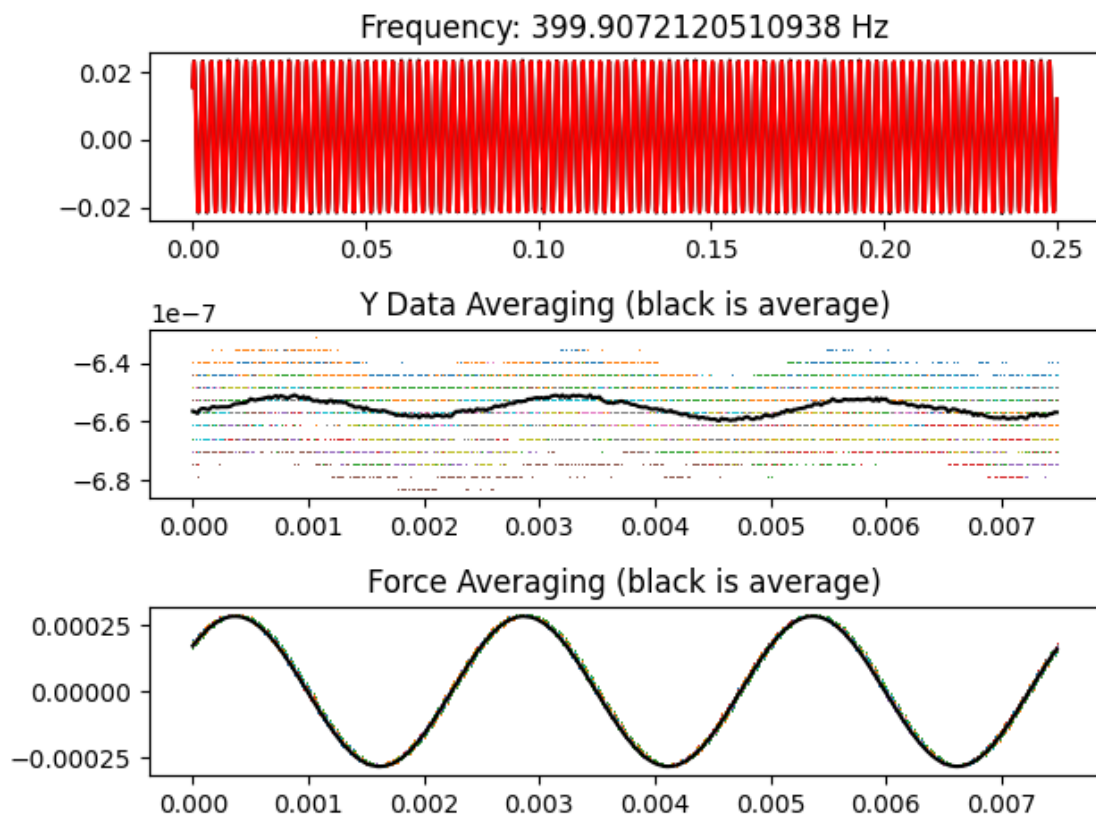


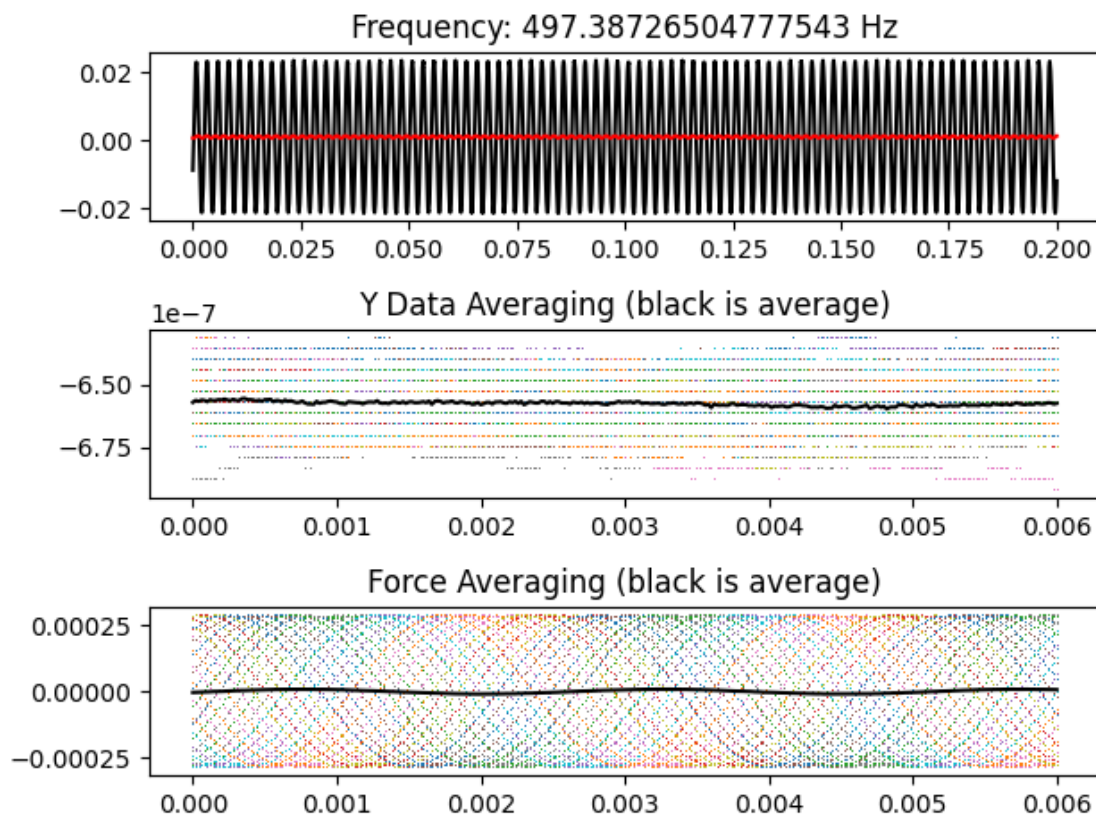


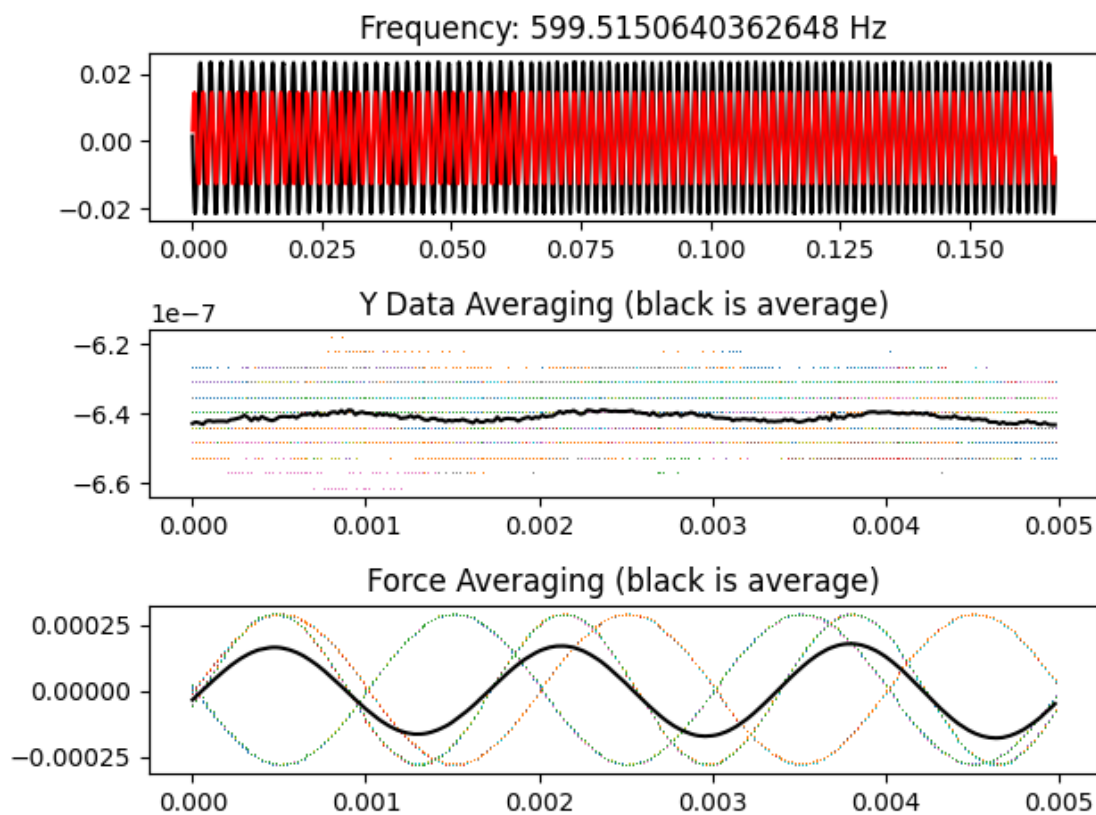


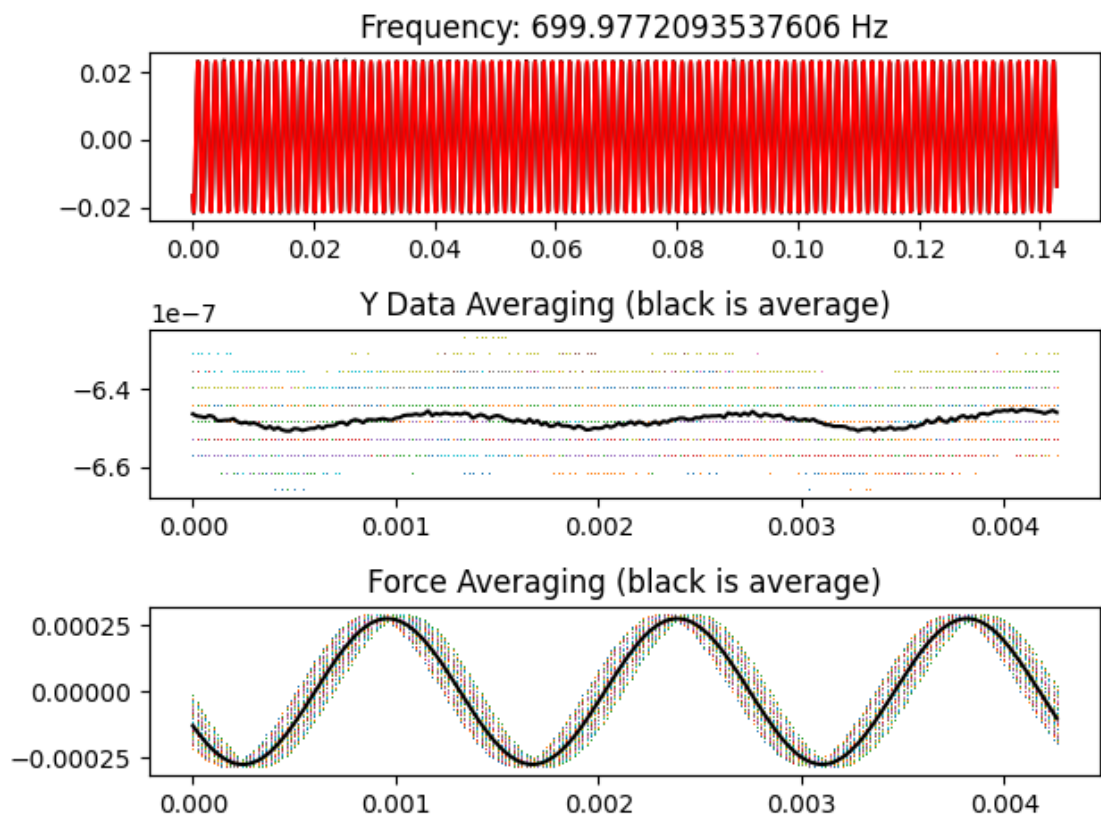


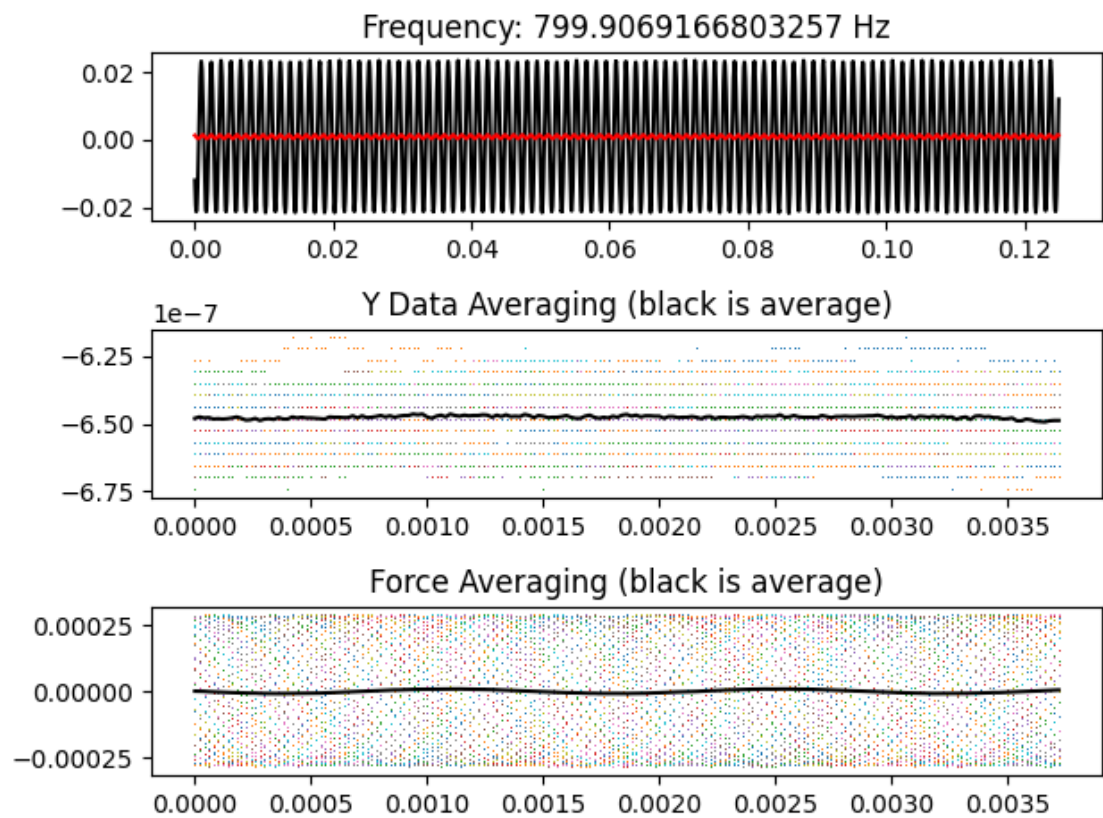


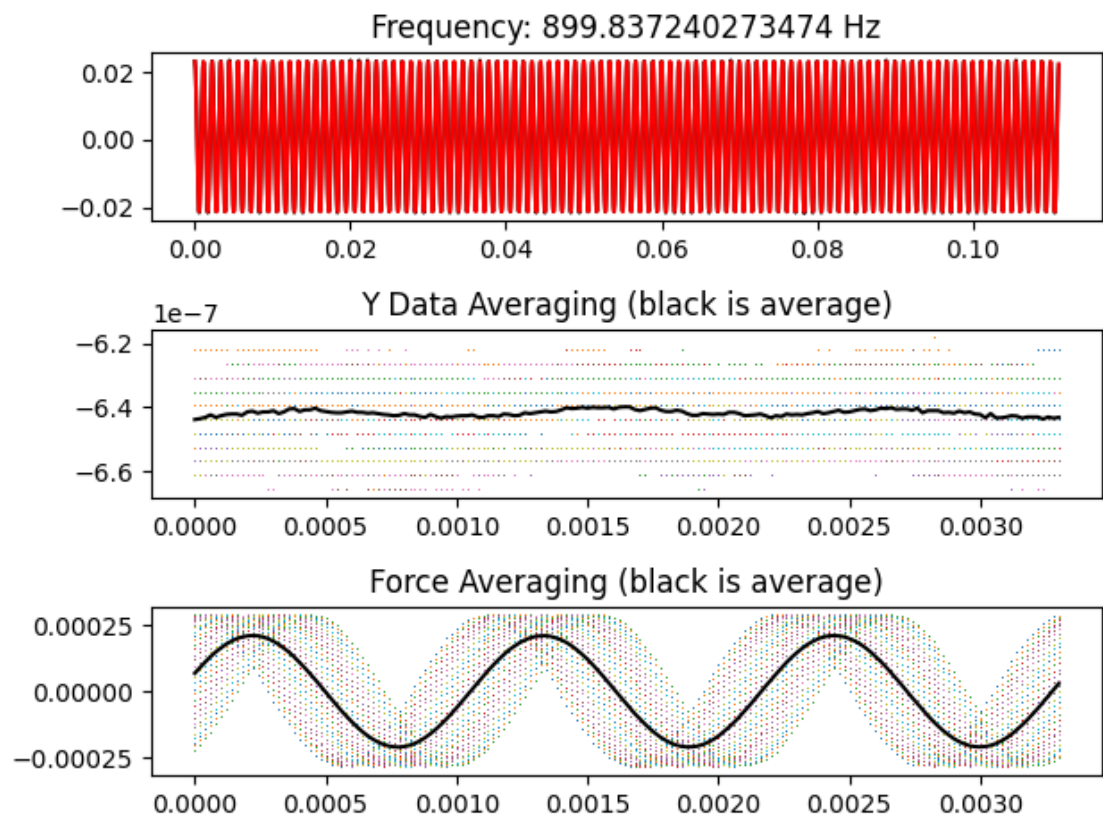


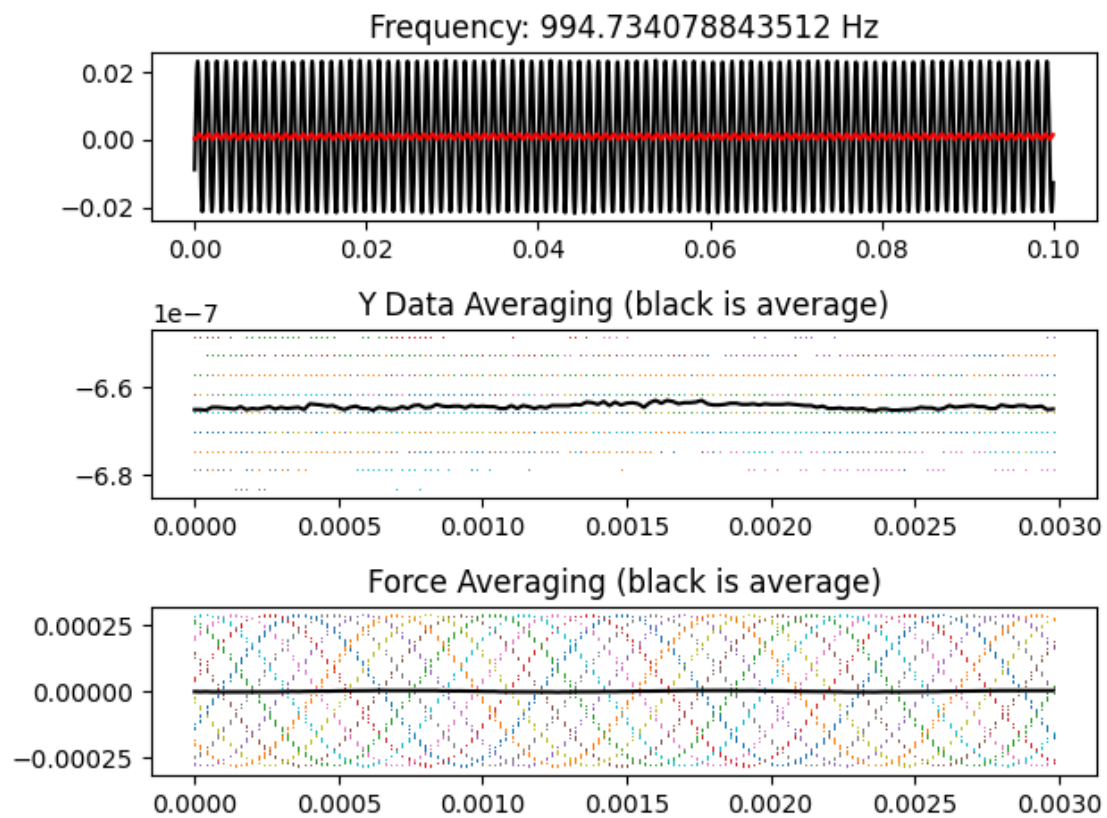


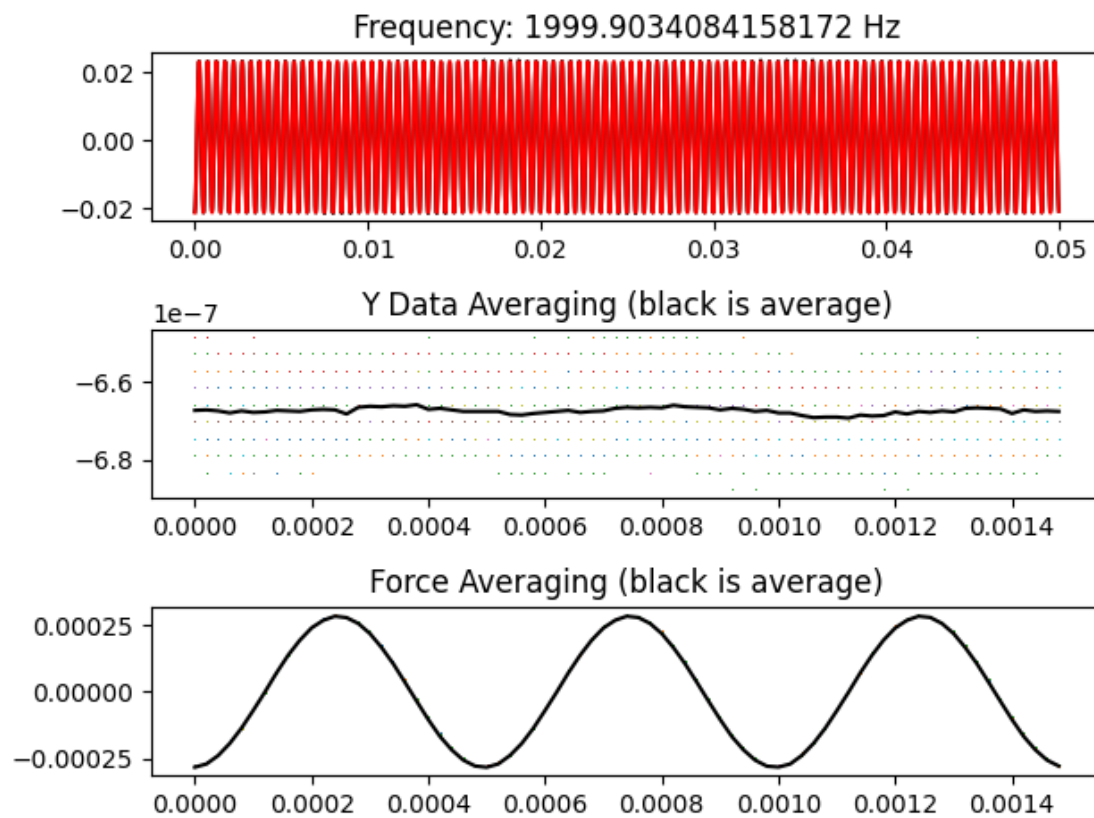


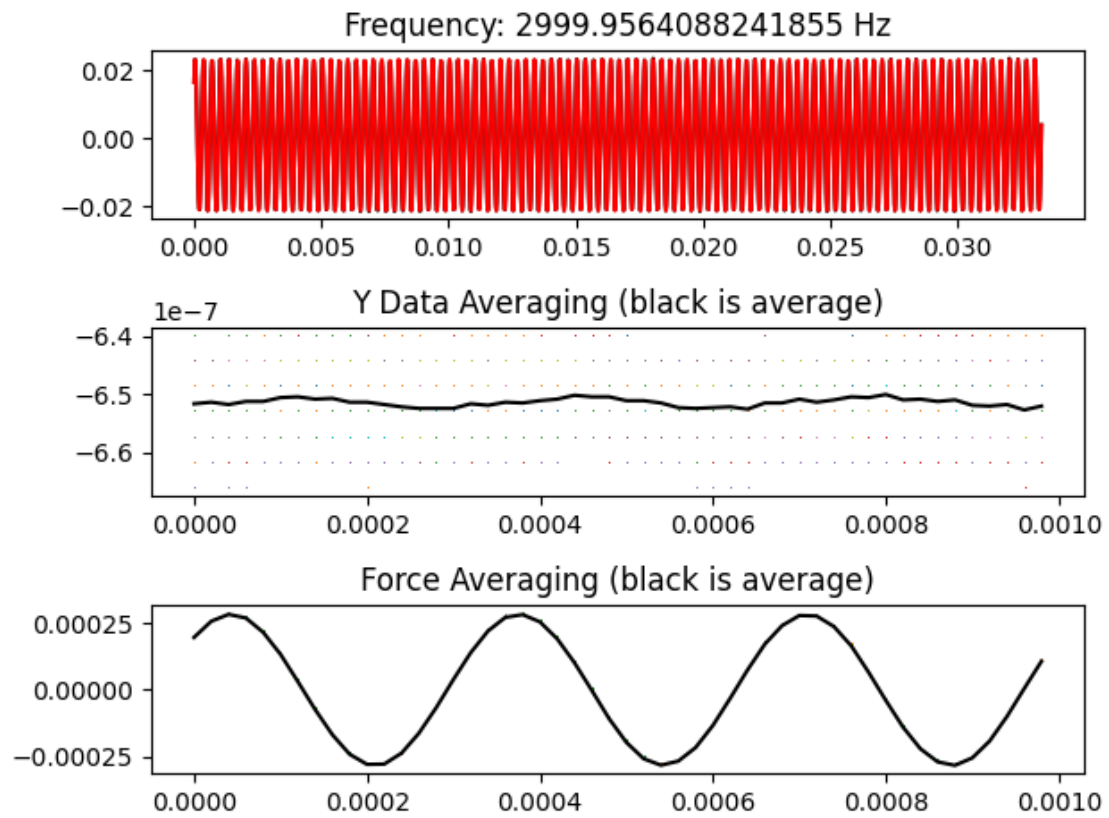


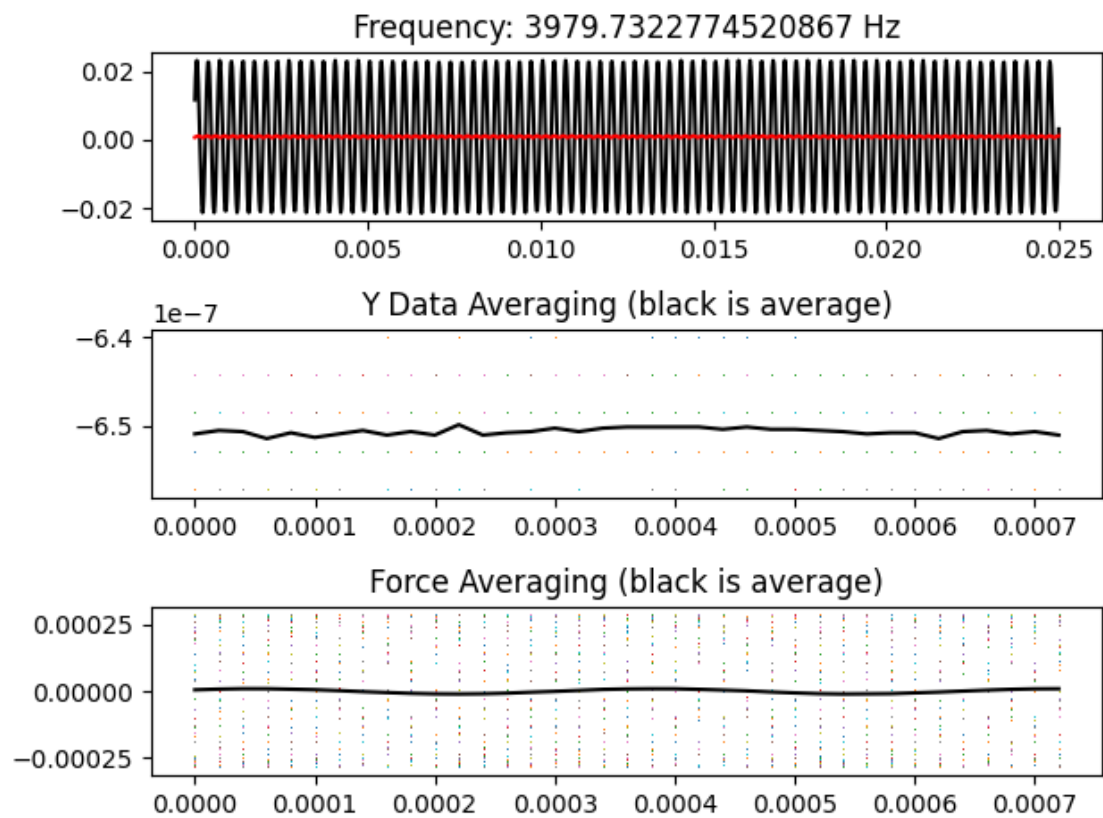


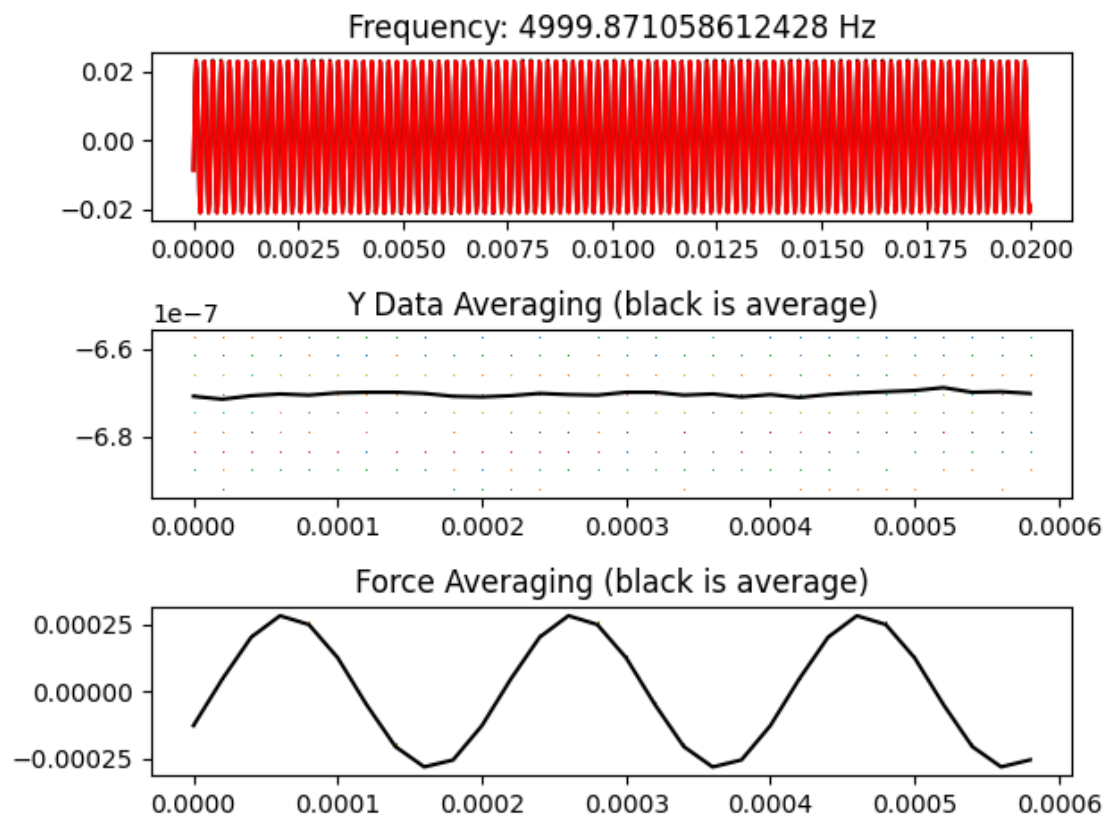


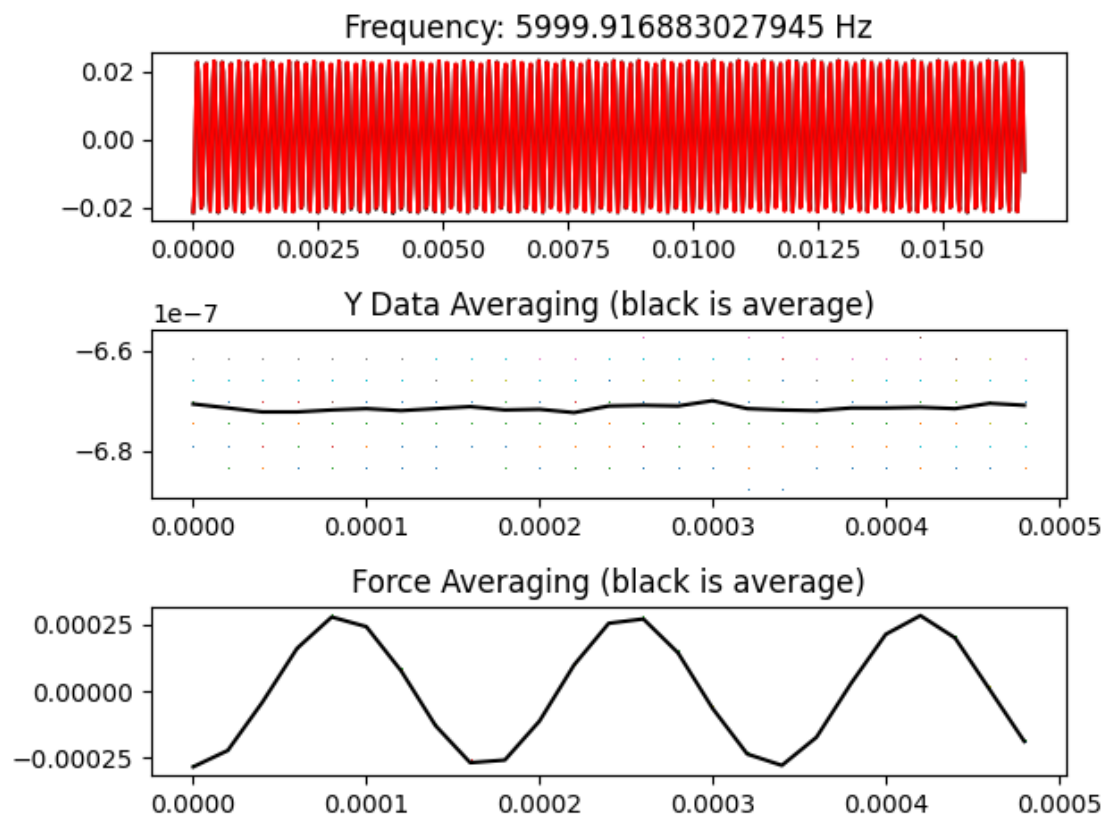


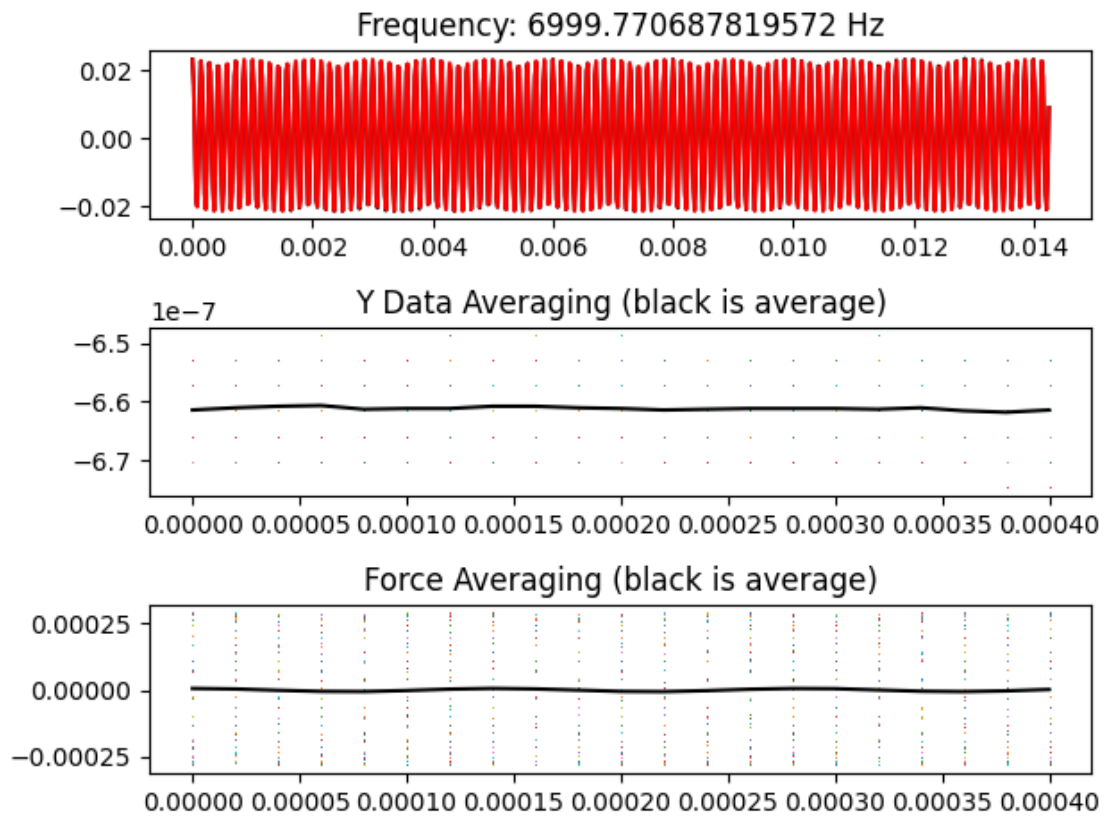


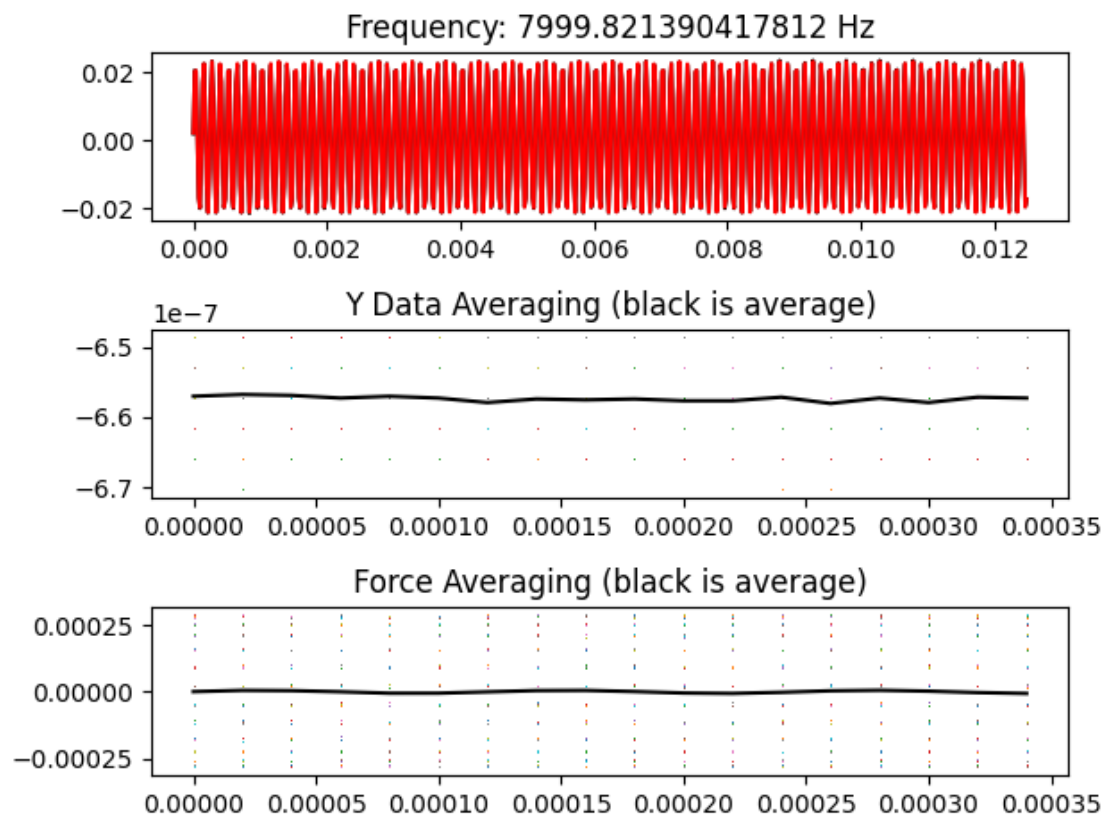


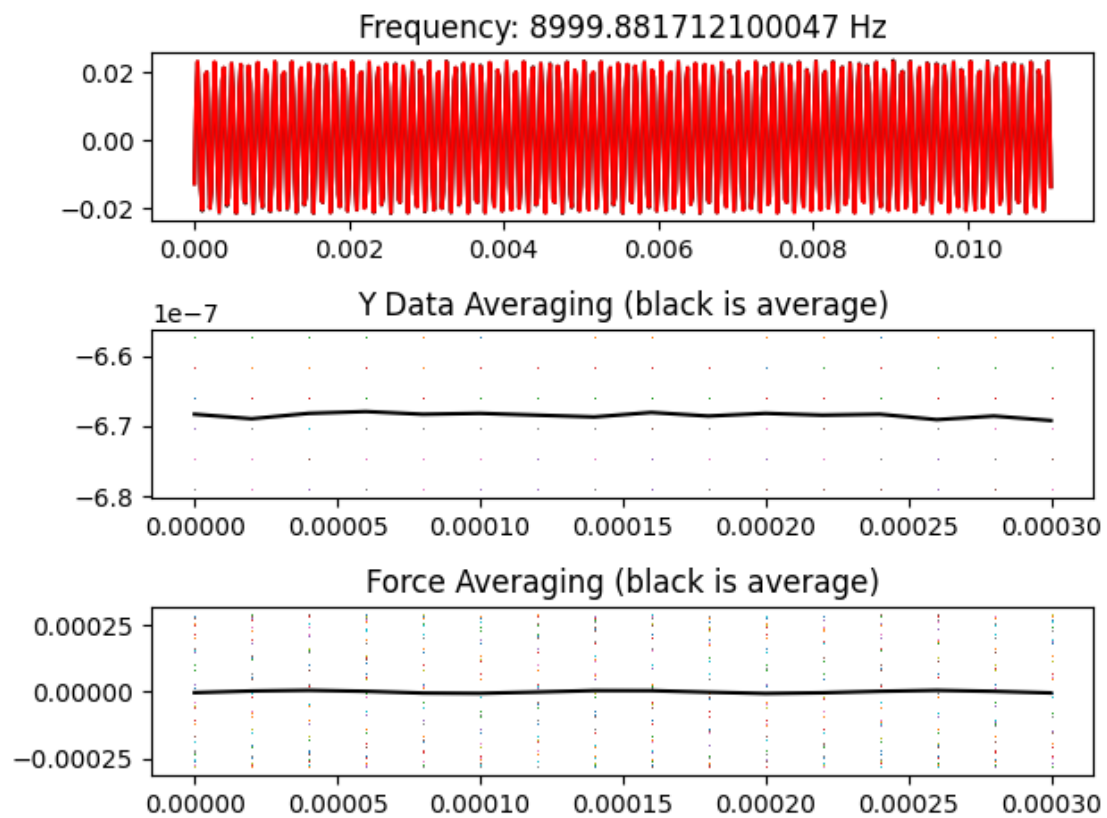


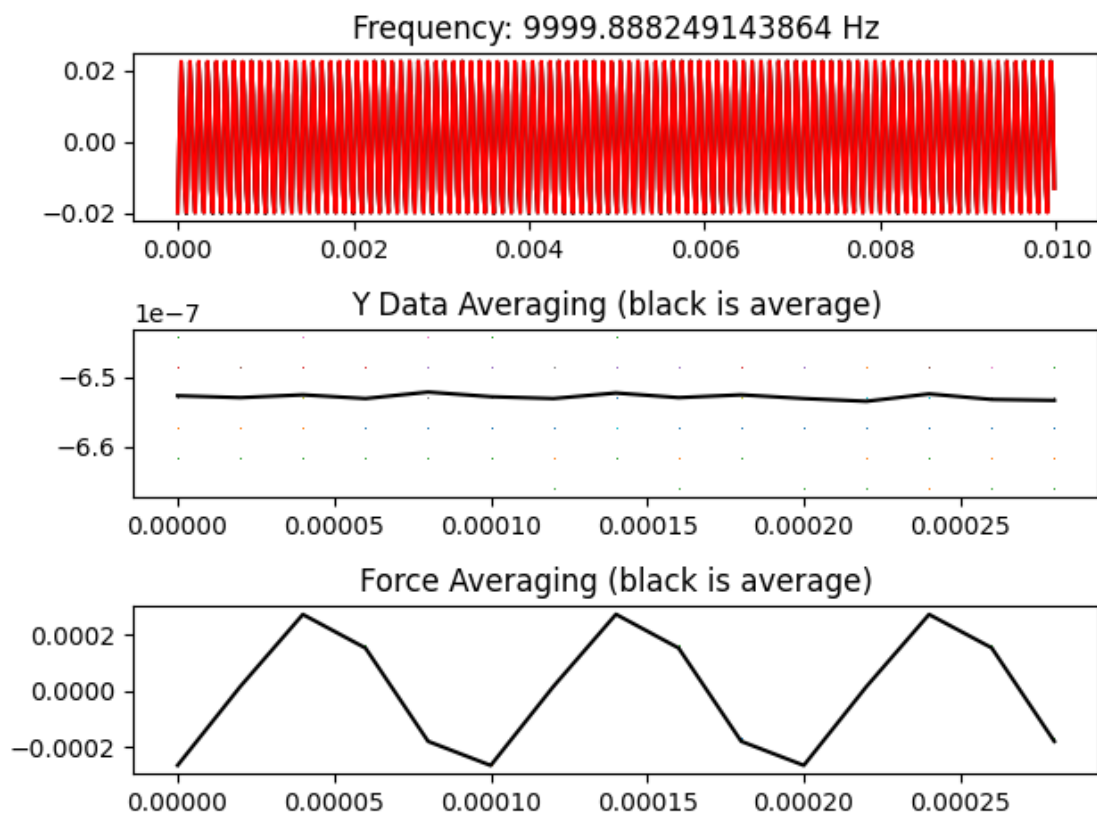


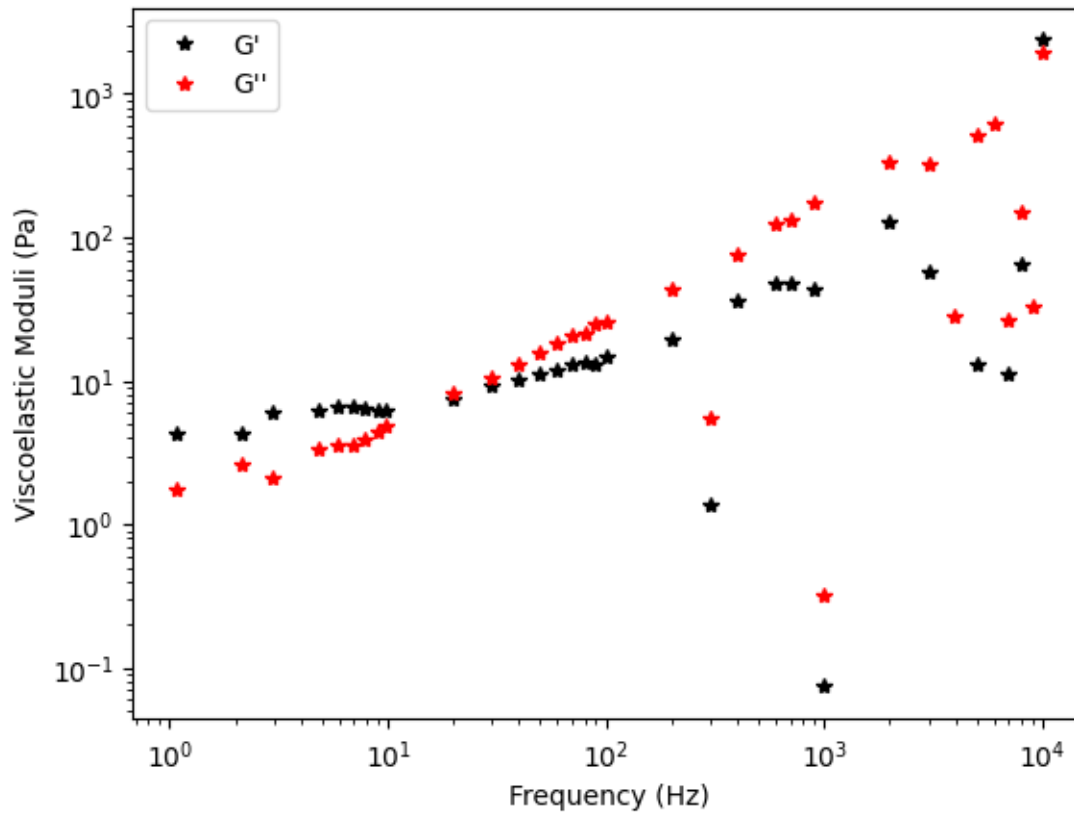












```
[20]: %reload_ext autoreload
      %autoreload 2
      from experiment import *
      params = [
          820074.2474561329,
          560741.2730406355,
          1.2912493813158651e-05,
          1.2801834934988684e-05,
          -7.858806662560769e-08,
          -6.621915318362614e-07
      ]
      datap = Passive_Analysis('250226/bead3p', params)
```

Calibration parameters:

Sx: 820074.2474561329

Sy: 560741.2730406355

kx: 1.2912493813158651e-05

ky: 1.2801834934988684e-05

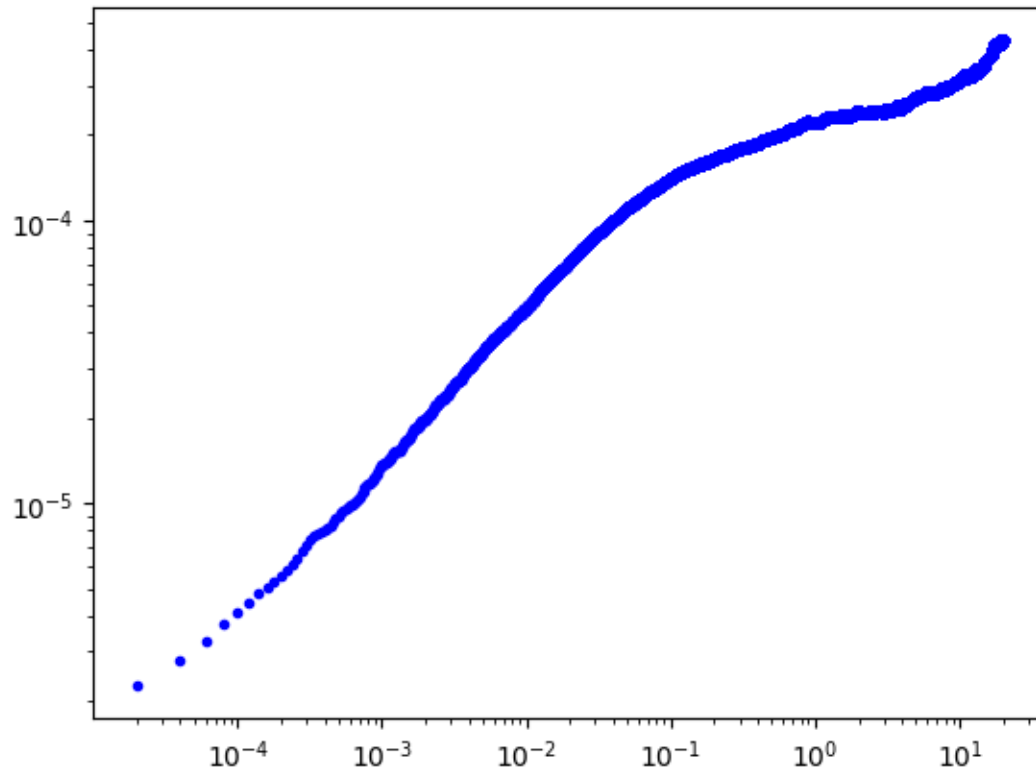
trapx: -7.858806662560769e-08

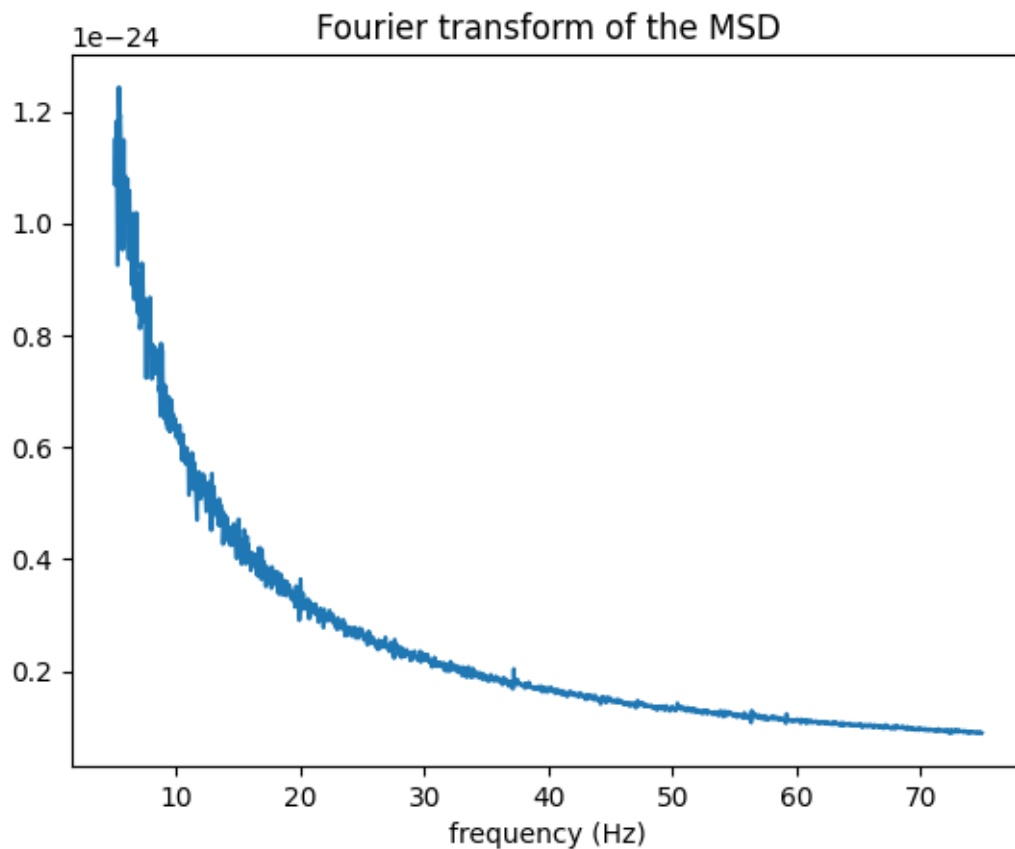
trapy: -6.621915318362614e-07

```
[21]: datap.final_analysis()
```

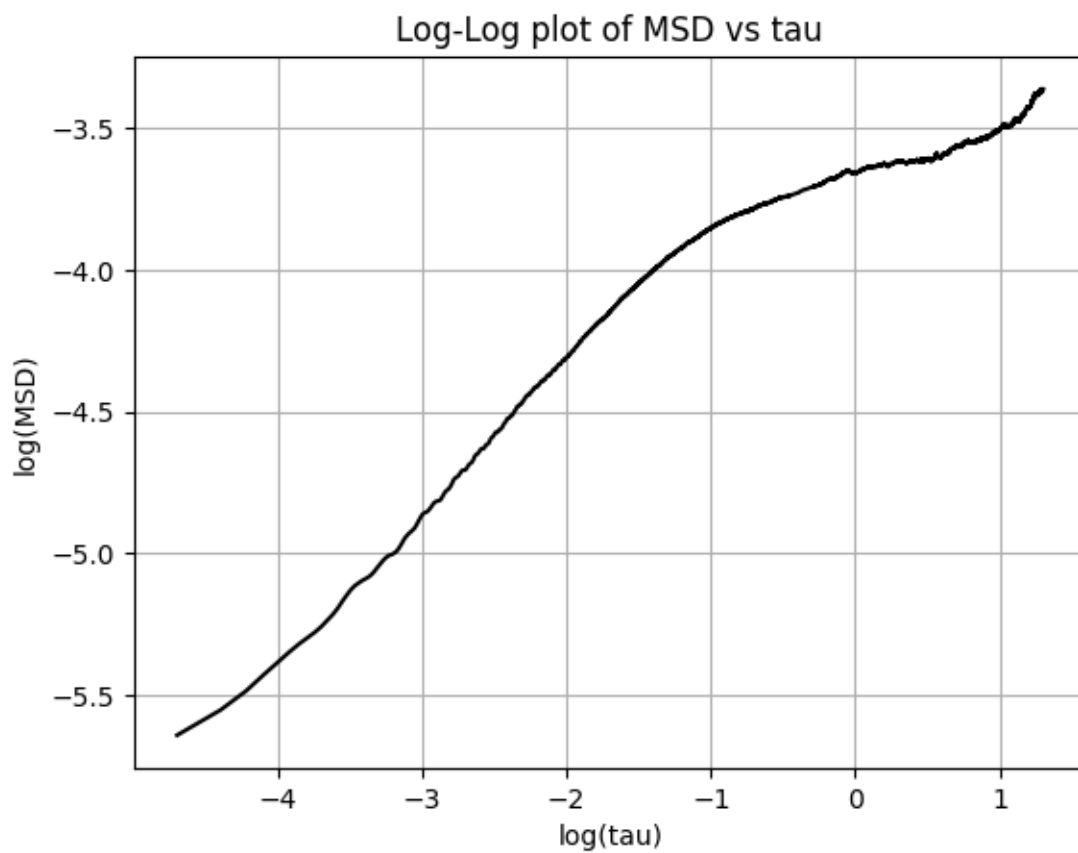
499999

999998





```
/Users/coledabois/.pyenv/versions/3.11.2/lib/python3.11/site-  
packages/pandas/core/arraylike.py:399: RuntimeWarning: divide by zero  
encountered in log10  
    result = getattr(ufunc, method)(*inputs, **kwargs)
```



Passive Analysis Parameters

